

Pruning Deep Neural Networks via the Marchenko–Pastur Distribution

Leonid Berlyand

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

Theo Bourdais

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Houman Owhadi

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Yitzchak Shmalo*

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

Abstract

We study a Marchenko–Pastur (MP) random-matrix approach to pruning deep neural networks. One theoretical component is a deterministic data-path certificate: if the removed component R has small propagated logit effect $L_s\|R\psi_1(s)\|_\infty$, then pruning R decreases an elastic-net objective and preserves any sample whose dense margin exceeds twice this perturbation. In the zero-budget limit this gives the “perfect pruning” statement: removing a component with no data-path logit effect does not reduce training-set accuracy. A prune–restore extension splits removed mass into a kept reservoir and a finally removed part, modeling restoration while holding the final sparse-execution pattern fixed. Separately, in a stylized additive model with L_2 regularization and one-sided stationarity, admissible random-like components must vanish along the training limit; persistent signal spikes then stabilize while the MP bulk collapses. MP enters as a sufficient condition: under iid-Gaussian admissible noise, the fitted edge σ_+ gives a high-probability layerwise budget signal.

On ImageNet-1k after three distillation epochs, ViT-B/16 2:4+ToMe reaches 83.41% top-1 (−1.70 pp from dense) at 59.81% sparse-execution MAC reduction, with a $1.365\times$ fixed-batch A40 native-2:4 backend speedup ($1.388\times$ best observed in the A40 batch sweep) for the same checkpoint and ToMe graph. A separate no-ToMe A100 endpoint comparison, not pooled with the ToMe-held-fixed result, gives $2.705\times$ measured speedup. At structured sparsity, ViT-B/16 6:12 reaches 83.74%, ViT-L/16 8:16 dense+permutation reaches 85.33% (−0.51 pp), and ConvNeXtV2-Base 12:16 reaches 86.35% (−0.37 pp). For CNNs, ResNet50 8:16 dense+permutation reaches 75.87% (−0.26 pp), and the ResNet152d CAST-conv+permutation row reaches 81.33% (−1.53 pp) at $\sim 50\%$ MAC accounting with a $1.62\times$ A40 im2col+2:4 sparse-GEMM audit. The wider 6:12, 8:16, and 12:16 rows are accuracy-focused $N:M$ projections rather than native 2:4 sparse-kernel throughput claims; the ResNet speed audit is an im2col endpoint, not a cuDNN Conv2d speedup.¹

Keywords: DNNs, ViTs, Random Matrix Theory, Marchenko–Pastur distribution, pruning, regularization

*Corresponding author: Yitzchak Shmalo. Email: yitzchak.shmalo@gmail.com.

¹Code: https://github.com/yspennstate/RMT_based_pruning_in_deep_learning. Companion methodology paper: https://github.com/yspennstate/RMT_based_pruning_in_deep_learning/blob/main/cast_2e_methodology.pdf. Checkpoints and per-run evidence files: <https://drive.google.com/drive/folders/1mm990SHAHL1YdISHxirvMRdVQEAjpIxDd>.

Norm convention. Throughout, $\|x\|_\infty$ denotes the vector maximum norm, while $\|A\|_\infty$ denotes the induced matrix norm

$$\|A\|_\infty := \max_i \sum_j |A_{ij}|.$$

Contents

1	Introduction	3
2	Randomness in Deep Neural Networks	4
2.1	Introduction to Deep Neural Networks	4
2.2	The MP distribution in machine learning contexts	4
2.3	Reduction of randomness in DNN weights via MP diagnostics	5
3	Numerical evidence: RMT-guided pruning of ViTs	6
3.1	ViT-B/16 results	6
3.2	Broader architecture sweep	6
3.3	Cert-aware structured sparsity: the FLOP / speedup table	6
3.4	Comparison with published structured-sparsity and ViT pruning baselines	6
3.5	Limitations and scope	8
4	Abstract Additive Perturbation Extensions	12
4.1	Assumptions for the generalized perturbation framework	12
4.2	Generalized perturbation and loss-reduction results	12
5	Main Theoretical Results: Deterministic Pruning Certificates	14
5.1	Deterministic data-path certificates for mask pruning	14
A	CAST/CAST-2E Recipe Table	24
B	Prior-Pruning Context	24
C	Theory–Algorithm–Result Mapping	25
D	Comparison Provenance and Scope Notes	25
E	Additional Deterministic Certificate Corollaries	26
F	Notes on the Abstract Additive Framework	28
F.1	Relocated abstract additive corollaries	28
G	Gaussian Specialization and Spectral Interpretation	30
G.1	Assumptions for the Gaussian specialization	30
G.2	Key perturbation lemma	32
G.3	A theorem on how removing randomness reduces the regularized loss	33
G.4	Asymptotic magnitude of admissible noise	36
H	Supplementary Numerical Results	40
H.1	Interpretation of the ViT-B/16 sweep	40
H.2	Multi-architecture cert connection	40

I	Proofs of the perturbation lemmas	41
I.1	Proofs of the deterministic pruning certificates	41
I.2	Proof of Lemma G.13	45
I.3	Proof of Lemma 4.4	45
I.4	Justification of Example F.6	46
I.5	Proof of Lemma G.15	46
I.6	Proof of Theorem G.16	46
I.7	Proof of Theorem G.20	47
J	Low-rank deformations and proofs of the main theorems	47
J.1	External random-matrix inputs quoted in the manuscript	47
J.2	Finite-rank deformations of random matrices	47
J.3	Proof of Proposition G.30	50
J.4	Proofs of Theorem G.18, Corollary G.19, Corollary G.14, and Theorem G.21 (Gaussian instances)	50
J.5	Proofs of the abstract-generalization corollaries and theorems	54
K	Algorithms for RMT-based pruning diagnostics	56
L	Spectral Edge Budgeting and hybrid RMT pruning protocols	57
M	CAST: certificate-aware 2:4 + ToMe conversion	57

1 Introduction

DNN compression is motivated by overfitting, regularization, and deployment constraints. Random Matrix Theory (RMT) has been used to study trained spectra, implicit self-regularization, generalization diagnostics, Jacobians, and initialization [36, 33, 38, 63, 58, 44, 45, 41, 37, 35]. For fixed pruning hyperparameters, this paper uses MP spectral diagnostics to allocate masks without validation or test access at mask-construction time.

The empirical target is sparsity inside dense affine maps of Vision Transformers and related ImageNet models [14, 60]. Trained spectra are heterogeneous: the method treats MP-like layers as candidates for larger pruning budgets, while protecting non-MP or heavy-tailed layers. The main numerical results are deliberately front-loaded. On ImageNet-1k, Hybrid Magnitude-SER keeps ViT-B/16 at 83.37% top-1 at 50% unstructured sparsity, while the deployable CAST 2:4+ToMe row reaches 83.41% after only three distillation epochs at 59.81% sparse-execution MAC reduction. The same checkpoint and ToMe graph gives a measured $1.365\times$ fixed-batch A40 native-2:4 speedup ($1.388\times$ best batch-sweep value); a separate no-ToMe ViT-B/16 dense-to-2:4 A100 endpoint gives $2.705\times$. Wider structured projections improve the accuracy side: ViT-B/16 6:12 reaches 83.74%, ViT-L/16 8:16 dense+permutation reaches 85.33% (-0.51 pp), and ConvNeXtV2-Base 12:16 reaches 86.35% (-0.37 pp). For CNNs, ResNet50 8:16 dense+permutation reaches 75.87%, only 0.26 pp below dense, and ResNet152d CAST-conv+permutation reaches 81.33%, 1.53 pp below dense, with a $1.62\times$ A40 im2col+2:4 sparse-GEMM audit. The wider 6:12, 8:16, and 12:16 rows are accuracy/MAC-accounting rows, not native sparse Tensor Core throughput claims.

The theoretical contribution has three headline pieces. First, the deterministic data-path certificate says that if the removed component R has small propagated logit effect $L_s\|R\psi_1(s)\|_\infty$, then pruning decreases an elastic-net objective and preserves every training sample whose dense margin is larger than twice this perturbation (Lemma 5.1, Corollary 5.2, Theorem 5.4). In the zero-budget or “perfect pruning” case, the margin bound gives no accuracy loss on the training set. Second, the prune-restore certificate

models structured $k:n$ sparsity: restoring entries inside an already-paid sparse-execution group can improve the certificate while leaving the final sparse-execution pattern unchanged (Theorem 5.5). Third, the additive L_2 -regularized theory shows that, under one-sided stationarity and local-path convergence, admissible random-like components vanish at the training limit; the associated MP bulk collapses while persistent signal spikes stabilize (Theorem G.21, Corollaries G.23–G.27, and the generalized Theorem 4.7). The MP edge enters as a sufficient random-matrix condition for data-path budgets and as the empirical layer-allocation signal used by SER/CAST. Table 6 gives literature context; Tables 1, 2, 3, 5, and 4 report the main unstructured, structured, projection-control, and deployability results. Prior pruning taxonomies and ViT/CNN compression citations are collected in Appendix B.

The theory supplies deterministic mask certificates and random sufficient conditions; these state the assumptions behind the pruning analysis rather than end-to-end correctness of the full ViT pipeline.

Section 2 gives DNN and MP preliminaries. Section 3 reports the numerical evidence. Sections 4 and 5 state the additive and deterministic certificates. Appendices contain provenance, proof, algorithm, and supplementary numerical material.

2 Randomness in Deep Neural Networks

2.1 Introduction to Deep Neural Networks

In classification tasks, the goal is to assign each element of a set S to one of K classes. Let $C(s) \in \{1, \dots, K\}$ denote the correct class of $s \in S$. Given a labeled training set $T \subset S$, we seek a classifier that generalizes from T to unseen data. We consider DNNs of the form

$$\phi(\cdot, \alpha) = \rho \circ X(\cdot, \alpha),$$

where ρ is the softmax map and $X(\cdot, \alpha)$ is a composition of affine maps and nonlinearities:

$$X(\cdot, \alpha) = \lambda \circ M_L(\cdot, \alpha) \circ \dots \circ \lambda \circ M_1(\cdot, \alpha).$$

Here:

- $M_k(\cdot, \alpha)$ is an affine map $\mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k}$ with weight matrix $W_k \in \mathbb{R}^{N_k \times N_{k-1}}$ and bias vector $\beta_k \in \mathbb{R}^{N_k}$, so $M_k(x) = W_k x + \beta_k$.
- $\lambda : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is a nonlinear activation. In the simplified theory below we take λ to be either the coordinatewise absolute value or ReLU.
- The softmax map $\rho : \mathbb{R}^K \rightarrow \mathbb{R}^K$ is given by

$$\rho(v)_i = \frac{e^{v_i}}{\sum_{j=1}^K e^{v_j}}, \quad v \in \mathbb{R}^K. \quad (1)$$

The standard cross-entropy loss is

$$L_{\text{CE}}(\alpha) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha)). \quad (2)$$

2.2 The MP distribution in machine learning contexts

The Marchenko–Pastur distribution is a basic object in RMT with applications in signal processing, statistics, wireless communication, and machine learning [61, 18, 48, 10]. We first define the relevant empirical spectral distributions.

Definition 2.1 (Eigenvalue and singular-value empirical spectral distributions). Let $G \in \mathbb{R}^{N \times M}$, and let $s_1(G), \dots, s_{\min\{N, M\}}(G)$ denote its singular values, counted with multiplicity and including possible zeros. The singular-value empirical spectral distribution (ESD) of G is

$$\nu_G := \frac{1}{\min\{N, M\}} \sum_{i=1}^{\min\{N, M\}} \delta_{s_i(G)}.$$

If $A \in \mathbb{R}^{M \times M}$ is symmetric positive semidefinite with eigenvalues $\lambda_1(A), \dots, \lambda_M(A)$, its eigenvalue ESD is

$$\mu_A := \frac{1}{M} \sum_{i=1}^M \delta_{\lambda_i(A)}.$$

Theorem 2.2 (Marchenko–Pastur law). *Let W_N be an $N \times M_N$ random matrix with i.i.d. entries of mean 0, variance σ^2 , and finite fourth moment. Define*

$$X_N := \frac{1}{N} W_N^\top W_N, \quad c_N := \frac{M_N}{N} \rightarrow c \in (0, \infty).$$

Then the eigenvalue ESD μ_{X_N} converges almost surely to the Marchenko–Pastur law

$$\mu_{\text{MP}}^{c, \sigma^2} = \left(1 - \frac{1}{c}\right)_+ \delta_0 + \frac{\sqrt{(\lambda_+ - x)(x - \lambda_-)}}{2\pi c \sigma^2 x} \mathbf{1}_{[\lambda_-, \lambda_+]}(x) dx, \quad (3)$$

where

$$\lambda_{\pm} = \sigma^2(1 \pm \sqrt{c})^2. \quad (4)$$

Proof. This is a classical cited random-matrix input rather than a new theorem of the paper; see Appendix J.1 for source notes.

Remark 2.3. If $M_N \leq N$ for all N , then $c \in (0, 1]$ and the atom at zero in (3) vanishes. When discussing the singular-value scale of W_N , we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding MP upper edge.

2.3 Reduction of randomness in DNN weights via MP diagnostics

At common initialization scale W_{ij} has variance g/N , so the remark above gives $\sigma_+ = \sqrt{g}(1 + \sqrt{c})$ in the rectangular aspect-ratio limit $M/N \rightarrow c$. Singular values and the nonzero eigenvalues of $W^\top W$ are then $O(1)$, while the eigenvalues of $X_\ell = W_\ell^\top W_\ell/N$ are $O(1/N)$; learned spectra later deviate from the initial random law [36, 52]. Training introduces structure, motivating the following signal-plus-randomness suppositions.

Supposition 1. After t training steps, the layer- ℓ weight matrix can be decomposed as

$$W_\ell(t) = R_\ell(t) + S_\ell(t),$$

where $R_\ell(t)$ is an independent random perturbation and $S_\ell(t)$ is a structured signal component.

Supposition 2. For the spectral spike interpretation, $S_\ell(t)$ is low rank or approximately low rank relative to the width.

These suppositions are modeling devices for the MP-budget argument, not assertions that trained ViTs are globally iid noise plus low-rank signal. Figure 1 gives two representative layer diagnostics; the assumptions behind using MP fit for budgets are stated in Sections 4 and 5.

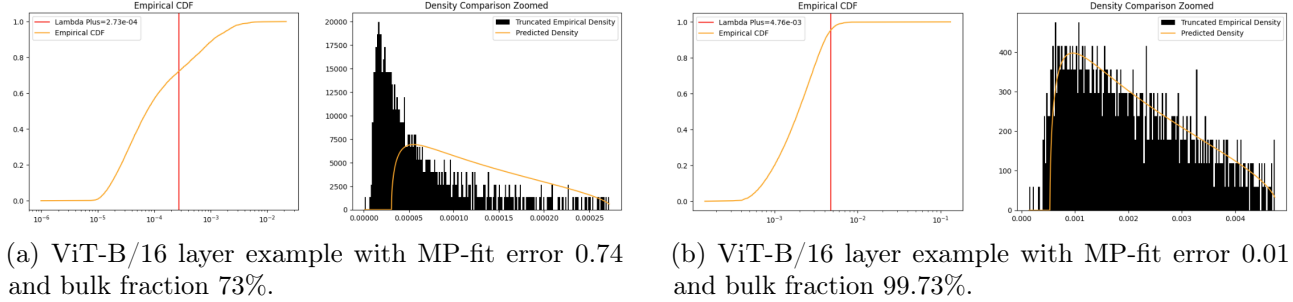


Figure 1: Layerwise MP diagnostics for two projection matrices from the same trained ViT-B/16.

3 Numerical evidence: RMT-guided pruning of ViTs

This section reports empirical pruning evidence; protocol/provenance details and the theory–algorithm mapping are in Appendices L and H.2 and the companion methodology paper.

3.1 ViT-B/16 results

Table 1: ViT-B/16 pruning results on ImageNet-1k validation under this paper’s matched checkpoint, prunable-layer set, sparsity grid, evaluation pipeline, and short fine-tuning budget. Entries are top-1 accuracy (%) at sparsity s ; the dense baseline is 85.11%. A dash denotes no completed full-validation checkpoint.

Method	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70
Classical magnitude	85.21	85.16	85.08	84.87	84.65	84.46	84.08	83.53	82.86	81.67	80.17	77.98	74.30	67.44
Classical RMT	85.21	85.14	85.01	84.97	84.77	84.55	84.31	83.98	83.31	82.57	81.31	79.13	76.46	71.53
SER	84.94	84.98	84.82	84.76	84.66	84.55	84.41	84.24	83.73	83.27	82.65	81.33	79.95	77.94
Hybrid Magnitude–SER	85.23	85.17	85.06	84.89	84.81	84.64	84.55	84.28	83.80	83.37	82.76	81.39	79.95	78.01

Layerwise interpretation and mask-only results are in Appendix H.1 and the companion methodology paper.

3.2 Broader architecture sweep

Table 2 gives the Hybrid Magnitude–SER sweep; Appendix H.2 and the methodology paper, quick-reference key `sec:seb-protocols` give architecture provenance, adaptive-variant details, slope/certificate-audit analyses, aggregate statistics, and checkpoint ledgers.

3.3 Cert-aware structured sparsity: the FLOP / speedup table

Table 3 summarizes structured sparsity; Algorithm 2 and source/checkpoint recipes are in Appendices M and A.

Deployability interpretation. Endpoint coverage and the separate A100 no-ToMe ViT-B benchmark are documented in Appendix D.

3.4 Comparison with published structured-sparsity and ViT pruning baselines

Table 6 is literature context, not matched baseline evidence; matched evidence is Table 1, the author-run AlphaPruning-style row, and the caveats in Appendix D. Table 7 gives the corresponding CNN pruning

Table 2: Hybrid Magnitude–SER results across ImageNet-1k checkpoints. Entries are post-FT top-1 accuracy (%) at sparsity s . “Base.” is the dense top-1 measured in this paper’s evaluation pipeline; ‡ marks the adaptive drop-threshold variant.

Architecture (timm checkpoint)	Base.	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70
ViT-B/16 augreg2_in21k_ft_in1k	85.11	85.23	85.17	85.06	84.89	84.81	84.64	84.55	84.28	83.80	83.37	82.76	81.39	79.95	78.01
ViT-B/16/384 augreg_in21k_ft	86.01	86.05	86.09	86.07	85.99	86.00	85.90	85.81	85.78	85.48	85.15	84.64	83.35	82.50	81.33
ViT-Large/16 augreg_in21k_ft	85.84	85.80	85.84	85.88	85.75	84.98	85.32	85.64	85.04	85.08	84.52	84.34	83.81	83.02	82.13
DeiT-Tiny patch16_224.fb_in1k	72.21	72.41	72.38	72.18	71.86	71.99	71.75	71.35	70.66	68.97	67.74	65.91	61.17	55.58	52.55
DeiT-Small patch16_224.fb_in1k	79.85	79.82	79.78	79.62	79.37	79.31	79.21	79.00	78.61	78.06	77.22	76.25	74.14	72.05	68.55
DeiT-Base patch16_224.fb_in1k	81.80	81.76	81.70	81.58	81.46	81.42	81.28	81.17	80.99	80.62	80.12	79.49	78.16	76.72	74.90
Swin-Tiny patch4_window7_224	81.20	81.32	81.28	81.25	81.05	81.09	80.91	80.78	80.37	80.14	79.80	78.98	78.06	76.64	74.47
ConvNeXt-Base in22k_ft_in1k	85.84	85.72	85.58	85.51	85.26	85.35	85.39	85.26	85.04	84.94	84.80	84.09	83.70	83.11	81.21
ResNet50d ra2_in1k	80.55	79.90	80.01	80.02	80.12	80.09	80.12	80.06	79.90	79.48	79.16	78.93	77.93	77.25	76.32
ResNet101d ra2_in1k	82.26	82.06	82.05	82.15	82.15	82.06	82.07	81.93	81.97	81.61	81.56	81.47	80.45	80.15	79.82
Hiera-Base+ mae_in1k_ft_in1k	84.40	85.04	84.94	84.76	84.39	77.95	83.12	82.15	82.37	82.29	82.25	82.00	80.03	80.55	78.96
ResNet18 tv_in1k‡	69.76	69.73	69.52	69.18	68.94	69.02	69.28	69.50	69.50	69.39	69.12	69.00	68.31	67.82	67.23
ResNet34 tv_in1k‡	73.28	73.00	72.67	72.39	72.62	73.06	73.09	73.16	73.16	73.00	72.88	72.74	72.12	71.59	71.44
ResNet50 tv_in1k‡	76.13	75.85	75.33	75.72	76.07	76.17	76.13	76.13	76.14	75.75	75.76	75.70	74.83	74.62	74.15
DeiT-Base patch16_224.fb_in1k‡	81.97	81.79	81.69	81.54	81.44	81.25	81.35	81.18	80.95	80.68	80.12	79.48	78.24	76.70	74.78
Swin-Tiny patch4_window7_224‡	81.39	81.32	81.29	81.26	81.05	80.77	80.47	80.58	80.43	80.11	79.80	78.94	78.03	76.53	74.31
ConvNeXtV2-Base fcmae_ft_in22k_in1k‡	86.72	86.67	86.58	86.43	86.27	85.93	85.97	85.96	85.71	85.58	85.33	84.81	84.61	83.83	81.40

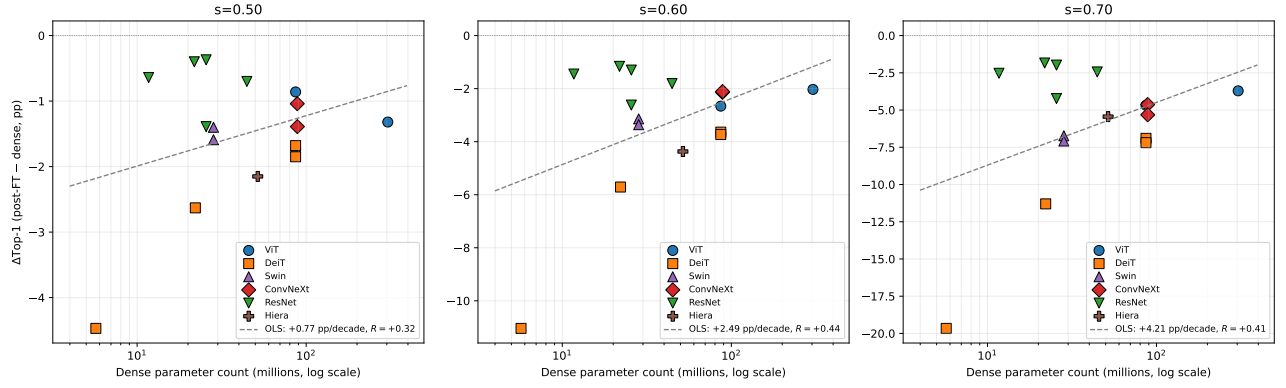


Figure 2: $\Delta\text{Top-1}$ relative to each dense baseline after Hybrid Magnitude–SER pruning vs. dense parameter count. Points are Table 2 entries; marker style encodes architecture family. Dashed lines are OLS fits at the plotted sparsity levels.

context.

Table 6: Selected ImageNet-1k pruning and structured-sparsity results. This is not an apples-to-apples ranking: rows start from different dense accuracies, checkpoints, architectures, compression axes, and fine-tuning budgets. Non-bold rows are cited results or author-run reimplementations; bold marks this paper’s rows only, not better performance. “FT/train” is the reported training or post-pruning fine-tuning budget.

Method	Architecture	Compression	Top-1	Δ vs. dense	FT/train	Notes
<i>Published ViT-family weight-pruning baselines</i>						
NViT [68]	DeiT-Base	global structured (34M params, 6.8G FLOPs)	83.29	−0.07	480 steps + 300 ep.	Hessian-aware structured pruning
SViT [7]	DeiT/SViT-Base	50% unstructured	81.51	−0.29	600 (train)	sparse-training from scratch
Spartan [57]	ViT-B/16	90% unstructured	81.18	+1.12‡	300 (train)	soft top- k + dual averaging
Hybrid Mag–SER	ViT-B/16	50% unstructured	83.37	−1.7	1/cycle	Table 2, $s = 0.50$
CAST 2:4+ToMe	ViT-B/16	2:4 + ToMe	83.41	−1.7	3	native 2:4 sparse-kernel path

Continued on next page

Table 6: Selected ImageNet-1k pruning and structured-sparsity results (continued).

Method	Architecture	Compression	Top-1	Δ vs. dense	FT/train	Notes
CAST 6:12 SER+α	ViT-B/16	6:12 = 50%	83.74	-1.4	3	ViT-B 6:12 structured row
CAST 8:16 dense+perm	ViT-L/16	8:16 = 50%	85.33	-0.51	3	dense-source wider-pattern row
<i>DeiT head-to-head (fine-tuned, reproduced by CP-ViT [50])</i>						
VTP [76]	DeiT-Base	projection-dim. pruning	80.70	-1.12	30	-43.20% FLOPs
PoWER-BERT [19]	DeiT-Base	token-vector elim.	80.17	-1.65	30	-39.24% FLOPs
HVT [42]	DeiT-Base	hierarchical pooling	79.94	-1.88	30	-44.78% FLOPs
CP-ViT [50]	DeiT-Base	cascade pruning	81.13	-0.69	30	-41.62% FLOPs (CP-ViT Table 4)
<i>Hardware-friendly structured-sparsity baselines and context rows</i>						
GPUSQ-ViT [70]	DeiT-Base + INT8	2:4 + 8-bit quant	82.90	+1.10	300 QAT/KD	-
GPUSQ-ViT [70]	DeiT-Base + INT4	2:4 + 4-bit quant	81.60	-0.20	300 QAT/KD	-
Beyond 2:4 [74]	DeiT-Base	V:N:M = 64:2:8 (300 ep)	81.08	-0.76	300	-
Beyond 2:4 [74]	DeiT-Base	V:N:M = 64:2:8 (600 ep)	81.76	-0.08	600 (train)	-
ELSA [23]	DeiT-Base	Uniform 2:4 (9.2G FLOPs)	81.80	+0.00	150 (train)	-
SERo [1]	DeiT-Base	sparse + re-optimization (5.4G FLOPs)	80.25	-1.55	200 (50 explore + 150 re-opt)	sparse exploration + re-optimization
CAST-conv+perm	ResNet152d	~50% MAC-accounting	81.33	-1.5	3	im2col audit; not cuDNN Conv2d speedup
CAST 8:16	ConvNeXt V2-B	50% MAC-accounting	85.85	-0.9	3	accuracy/MAC probe; no native 8:16 kernel
CAST 12:16	ConvNeXt V2-B	~25% MAC-accounting	86.35	-0.4	3	accuracy/MAC probe; no native 12:16 kernel
<i>Author-run baseline on this paper's protocol</i>						
AlphaPruning-style (ours) [32]	ViT-B/16	50% layerwise- α + short FT	~ 82.9	~ -2.2	~ 3	layerwise α allocation

[†]Spartan reports Δ against its own dense baseline; see Appendix D for row provenance and author-run baseline notes.

3.5 Limitations and scope

The main empirical scope qualifications are wider-pattern deployability, ResNet Conv2d endpoint support, artifact-bundle coverage, matched-ablation scope beyond ViT-B/16, and single-seed CAST row estimates; details are in Appendix D. The main theory limitation is the unestimated analytic L_s for modern transformer blocks; see Remark F.9.

Table 3: CAST $k:n$ structured-projection results. “MAC red.” is dense-equivalent MAC-accounting reduction; for rows without a deploy-speed entry this is not an audited sparse-backend wall-clock claim. “Th. spd.” is $1/(1 - \text{MAC red.})$, and Δ is top-1 minus the corresponding dense baseline. “Deploy spd.” is a measured sparse-backend speedup only when listed; Table 4 reports endpoint, batch, and best-observed audit details. A dash means no exact sparse-backend wall-clock speedup is claimed.

Architecture	Source s	Method	Dense MACs	MAC red.	Th. spd.	Deploy spd.	Top-1 (%)	Δ (pp)
<i>ViT family</i>								
ViT-B/16	0.35	Magnitude 2:4 + ToMe	17.56G	59.81%	2.49×	1.36×	82.92	−2.19
ViT-B/16	0.35	CAST 2:4 + ToMe	17.56G	59.81%	2.49×	1.36×	83.41	−1.70
ViT-B/16	0.35	CAST 6:12 SER + $\alpha=0.5$ (no ToMe)	17.56G	50%	2.00×	—	83.74 [§]	−1.37
ViT-L/16	0.35	CAST 2:4 + ToMe	61.55G [†]	~60% [†]	2.50×	1.37×	84.37	−1.47
ViT-L/16	dense	CAST 8:16 dense+perm (no ToMe)	61.55G	50%	2.00×	—	85.33 [§]	−0.51
DeiT-B	0.35	CAST 2:4 + ToMe	17.56G	59.81%	2.49×	1.36×	80.48	−1.32
DeiT-S	0.35	CAST 2:4 + ToMe	4.61G	59.81%	2.49×	1.38×	76.96	−2.89
DeiT-T	0.35	CAST 2:4 + ToMe	1.26G	59.81%	2.49×	1.33×	65.93	−6.28
<i>ResNet family</i>								
ResNet50	0.35	CAST-conv	4.09G	48.5%	1.94×	—	73.14	−2.99
ResNet50	0.35	CAST-conv+perm	4.09G	48.5%	1.94×	1.70×	75.67 [‡]	−0.46
ResNet50	dense	CAST 8:16 dense+perm	4.09G	50%	2.00×	—	75.87 [§]	−0.26
ResNet50d	0.35	CAST-conv	4.33G	49.85%	1.99×	—	78.08	−2.47
ResNet50d	0.35	CAST-conv+perm	4.33G	49.85%	1.99×	1.50×	78.00 [‡]	−2.55
ResNet50d	dense	CAST 8:16 dense+perm	4.33G	50%	2.00×	—	78.57 [§]	−1.98
ResNet101d	0.35	CAST-conv	8.0G	~50%	2.00×	—	80.13	−2.13
ResNet101d	0.35	CAST-conv+perm	8.0G	~50%	2.00×	1.57×	80.59 [‡]	−1.67
ResNet101d	dense	CAST 8:16 dense+perm	8.0G	50%	2.00×	—	80.92 [§]	−1.34
ResNet152d	0.35	CAST-conv+perm	11.8G	~50%	2.00×	1.62×	81.33 [‡]	−1.53
<i>ConvNeXt family</i>								
ConvNeXtV2-Base	0.35	CAST 2:4 cert + free-restore	15.4G	50%	2.00×	1.26×	85.47	−1.25
ConvNeXtV2-Base	dense	CAST 12:16 dense+perm (25% sparse)	15.4G	25%	1.33×	—	86.35 [§]	−0.37
ConvNeXtV2-Base	dense	CAST 8:16 dense+perm (50% sparse)	15.4G	50%	2.00×	—	85.85 [§]	−0.87

[†]Analytical ViT-L/16 ToMe+2:4 estimate. [‡]Exact im2col+2:4 sparse-GEMM audit, not faster cuDNN Conv2d.

[§]Accuracy/MAC-accounting probe unless a measured endpoint is listed; dense-source wider-pattern rows are not incremental comparisons to the $s = 0.35$ source rows. *Magnitude row shares the ViT-B 2:4+ToMe deployment path; see Appendix D.

Table 4: Deployability audit for the main measured checkpoints, reporting best-observed A40 batch-sweep values. Accuracy is fixed by the saved post-FT checkpoint; no weights are rewritten, and only the in-memory execution endpoint changes. Linear rows use the PyTorch/NVIDIA native 2:4 backend on A40. ResNet rows use exact im2col+2:4 sparse GEMM relative to dense im2col; these rows do not claim a faster cuDNN Conv2d endpoint.

Checkpoint	Audited endpoint	Batch	Dense im/s	Sparse im/s	Speedup	Status
ViT-B/16 + ToMe	native Linear 2:4	64	1191.8	1654.5	1.388×	native sparse Tensor Core path
ViT-L/16 + ToMe	native Linear 2:4	64	620.8	865.6	1.394×	native sparse Tensor Core path
DeiT-B + ToMe	native Linear 2:4	64	1189.6	1646.7	1.384×	native sparse Tensor Core path
DeiT-S + ToMe	native Linear 2:4	128	2708.5	3725.8	1.376×	native sparse Tensor Core path
DeiT-T + ToMe	native Linear 2:4	128	5158.4	6859.4	1.330×	native sparse Tensor Core path
ConvNeXtV2-B	native Linear 2:4	64	499.3	646.5	1.295×	pointwise Linear layers accelerated; dense convs unchanged
ResNet50	im2col sparse GEMM	128	469.5	797.9	1.700×	im2col 2:4 sparse GEMM
ResNet50d	im2col sparse GEMM	128	445.2	666.2	1.496×	im2col 2:4 sparse GEMM
ResNet101d	im2col sparse GEMM	128	228.2	357.8	1.568×	im2col 2:4 sparse GEMM
ResNet152d	im2col sparse GEMM	128	159.9	258.7	1.617×	im2col 2:4 sparse GEMM

Table 5: Projection controls inside this paper. Each row compares an original/control projection to the later projection under the same accuracy convention as Table 3. “FT ep.” is the post-projection distillation budget for the two rows being compared. Wider dense-source rows are accuracy/MAC probes and should not be read as native sparse-kernel speedups.

Architecture	Original/control projection	Compared projection	pro- FT ep.	Top-1 change	MAC/speed change
ViT-B/16	Magnitude 2:4+ToMe	CAST 2:4+ToMe	3 → 3	82.92 → 83.41 (+0.49)	same 59.81% MAC red.; same native 2:4 path
ViT-B/16	CAST 2:4+ToMe	CAST 6:12 ToMe	no 3 → 3	83.41 → 83.74 (+0.33)	59.81% → 50% MAC red.; no native 6:12 audit
ViT-L/16	CAST 2:4+ToMe	CAST dense+perm	8:16 3 → 3	84.37 → 85.33 (+0.96)	~ 60% → 50% MAC red.; no native 8:16 audit
ResNet50	CAST-conv	CAST-conv+perm	3 → 3	73.14 → 75.67 (+2.53)	same 48.5% MAC red.; 1.70× im2col audit
ResNet50	CAST-conv+perm	CAST dense+perm	8:16 3 → 3	75.67 → 75.87 (+0.20)	48.5% → 50% MAC red.; no native 8:16 audit
ResNet50d	CAST-conv	CAST-conv+perm	3 → 3	78.08 → 78.00 (−0.08)	same 49.85% MAC red.; 1.50× im2col audit
ResNet50d	CAST-conv+perm	CAST dense+perm	8:16 3 → 3	78.00 → 78.57 (+0.57)	49.85% → 50% MAC red.; no native 8:16 audit
ResNet101d	CAST-conv	CAST-conv+perm	3 → 3	80.13 → 80.59 (+0.46)	same ~ 50% MAC red.; 1.57× im2col audit
ResNet101d	CAST-conv+perm	CAST dense+perm	8:16 3 → 3	80.59 → 80.92 (+0.33)	same 50% MAC red.; no native 8:16 audit
ConvNeXtV2-B	CAST cert+free	2:4 CAST dense+perm	8:16 3 → 3	85.47 → 85.85 (+0.38)	same 50% MAC red.; no native 8:16 audit
ConvNeXtV2-B	CAST cert+free	2:4 CAST dense+perm	12:16 3 → 3	85.47 → 86.35 (+0.88)	50% → 25% MAC red.; no native 12:16 audit

Table 7: CNN ImageNet pruning context. Positive “drop” means lower top-1 than the row’s dense baseline; negative drop means the pruned row is above that baseline. External rows use each cited paper’s checkpoint, pruning axis, and training recipe, so the table is context rather than a matched leaderboard. “FT/train” is the reported post-pruning fine-tuning or training budget; n/r means the cited table/protocol does not give a single epoch count for that row.

Method	Model	Compute red.	Top-1	Drop	FT/train	Speed/deploy	note
SFP [20]	ResNet-50	41.8% FLOPs	74.61	1.54	n/r		filter/channel-pruned CNN
HRank [31]	ResNet-50	43.8% FLOPs	74.98	1.17	n/r		feature-map-rank filter pruning
FPGM [21]	ResNet-50	53.5% FLOPs	74.83	1.32	n/r		geometric-median filter pruning
ResRep [13]	ResNet-50	54.5% FLOPs	76.15	0.00	180 train		structural re-parameterization; narrowed dense CNN
DepGraph [15]	ResNet-50	51.8% MACs	75.83	0.32	n/r		automatic structural pruning
Isomorphic [16]	ResNet-50	50.1% MACs	75.91	0.22	300 FT		structural pruning; cited paper reports real latency caveats
Isomorphic [16]	ResNet-101	50.9% MACs	77.43	-0.05	300 FT		structural pruning, above its dense baseline
Isomorphic [16]	ResNet-152	64.9% MACs	77.84	0.47	300 FT		structural pruning with narrowed dense kernels
Isomorphic [16]	ConvNeXt-B→S-width	44.8% MACs	83.17	0.66	300 FT		pruned ConvNeXt-style dense model
Isomorphic [16]	ConvNeXt-B→T-width	72.7% MACs	82.19	1.64	300 FT		pruned ConvNeXt-style dense model
CAST-conv+perm	ResNet50	48.5% MACs	75.67	0.46	3		1.70× A40 im2col+2:4 audit; not cuDNN Conv2d
CAST 8:16 dense+perm	ResNet50	50.0% MACs	75.87	0.26	3		no native 8:16 sparse-kernel claim
CAST-conv+perm	ResNet152d	~ 50% MACs	81.33	1.53	3		1.62× A40 im2col+2:4 audit; stronger timm dense baseline
CAST 8:16 dense+perm	ConvNeXtV2-B	50.0% MACs	85.85	0.87	3		no native 8:16 sparse-kernel claim
CAST 12:16 dense+perm	ConvNeXtV2-B	25.0% MACs	86.35	0.37	3		no native 12:16 sparse-kernel claim

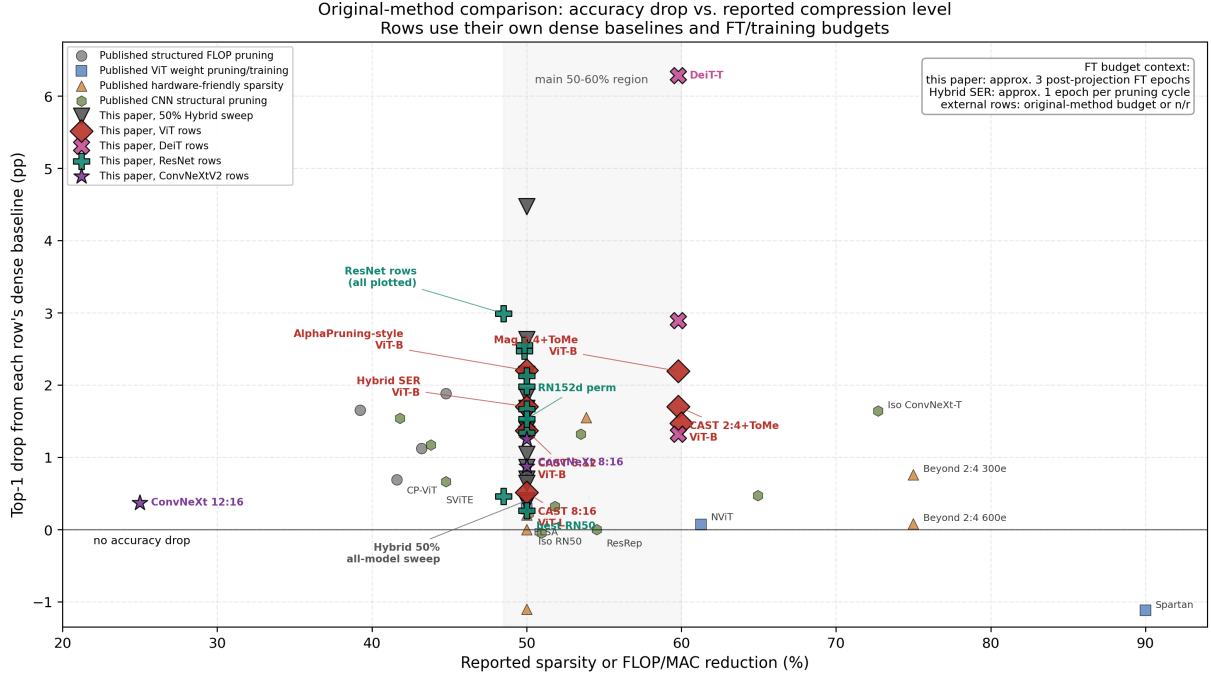


Figure 3: Descriptive comparison against the original methods under their own protocols. Each external point uses the cited paper’s dense baseline, compression axis, and reported FT/training budget; this paper’s larger markers use the budgets in Tables 6 and 7 (3 post-projection epochs for CAST rows and about one epoch per Hybrid pruning cycle). The y-axis is the drop from that row’s own dense baseline, so lower is a smaller accuracy loss and negative values are above the row’s dense baseline. Because checkpoints, pruning axes, and FT budgets differ, the figure is provenance-normalized context rather than a matched leaderboard.

4 Abstract Additive Perturbation Extensions

This section gives the abstract $W = S + R$ additive analogue of the deterministic data-path framework in Section 5. Gaussian/RMT specializations and proofs are in Appendices G, F, I, and J.5.

4.1 Assumptions for the generalized perturbation framework

Throughout the asymptotic statements in this section, the training set T is finite and fixed independently of N , unless a result explicitly states summability conditions for a growing family.

Assumption 4.1 (General architecture). Assume the DNN can be written as

$$\phi = \rho \circ \psi_2 \circ (R + S) \circ \psi_1,$$

where ψ_1, ψ_2 may depend on the remaining network parameters and ρ is softmax. For each $s \in T$, let $\mathcal{U}_s$ be any set containing the admissible interpolation segment $\{(S + \vartheta R)\psi_1(s) : \vartheta \in [0, 1]\}$. We assume that ψ_2 has a finite local Lipschitz constant with respect to ℓ_∞ on $\mathcal{U}_s$:

$$L_{\psi_2}(s) := \sup_{\substack{v, w \in \mathcal{U}_s \\ v \neq w}} \frac{\|\psi_2(v) - \psi_2(w)\|_\infty}{\|v - w\|_\infty} < \infty.$$

Assumption 4.2 (Perturbative random component). Assume that for every v measurable with respect to the conditioning variables $\sigma(S, \psi_1, \psi_2)$, including deterministic v ,

$$\mathbb{P}\left(\|Rv\|_\infty > d_1(N)J(v) \mid S, \psi_1, \psi_2\right) \leq d_2(N), \quad (5)$$

where $d_1(N), d_2(N) \rightarrow 0$ and $J(v)$ depends only on v . When a spectral interpretation is desired, we additionally assume that the singular-value ESD of R converges to a deterministic law with finite right edge $b_+ < \infty$.

Assumption 4.3 (Structured component for spectral conclusions). For any spectral conclusion, assume that there exists a sequence $k_N = o(\min\{N, M\})$ such that

$$\sum_{i > k_N} s_i(S)^2 \rightarrow 0.$$

The convergence is in probability when S is random.

For a DNN satisfying Assumption 4.1, define

$$h_\phi(s) := J(\psi_1(s)) L_{\psi_2}(s), \quad a_\phi(N, s) := h_\phi(s) d_1(N). \quad (6)$$

4.2 Generalized perturbation and loss-reduction results

We consider the regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha(t))) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i(t)\|_F^2. \quad (7)$$

Lemma 4.4. *Let*

$$X = \psi_2 \circ (R + S) \circ \psi_1(s),$$

and let α_W, α_S denote the corresponding network parameters when the relevant weight matrix is $W = R + S$ and S , respectively. Under Assumptions 4.1–4.2,

$$\mathbb{P}\left(\|X(s, \alpha_S) - X(s, \alpha_W)\|_\infty \leq d_1(N)h_\phi(s)\right) \geq 1 - d_2(N).$$

Theorem 4.5. Assume Assumptions 4.1–4.2 and that

$$\langle S, R \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Assume further that for all $s \in T$,

$$a_\phi(N, s) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Then removing R yields

$$L(\alpha_S) = L(\alpha_W) - \lambda_{\text{reg}} \|R\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

Secondary loss and finite-rank deformation corollaries are stated in Appendix F.1.

Theorem 4.6 (Multi-layer telescoping loss reduction). Let $\mathcal{P}$ be a finite set of layers selected for pruning, with $|\mathcal{P}|$ fixed independently of N . For each $\ell \in \mathcal{P}$, write

$$W_\ell = S_\ell + R_\ell,$$

and assume that, after the previous layers in $\mathcal{P}$ have already been replaced by their structured parts, the single-layer hypotheses of Theorem 4.5 continue to hold for layer ℓ with perturbation scale $a_{\phi, \ell}(N, s)$. If $\alpha^{(0)}$ denotes the original network parameters and $\alpha^{(|\mathcal{P}|)}$ the parameters after replacing every W_ℓ by S_ℓ , then

$$L(\alpha^{(|\mathcal{P}|)}) = L(\alpha^{(0)}) - \lambda_{\text{reg}} \sum_{\ell \in \mathcal{P}} \|R_\ell\|_F^2 + o_{\mathbb{P}}(1).$$

More generally, the set of pruned layers may depend on N . Let $\mathcal{P}_N$ be a sequence of layer sets, ordered in the pruning order. Assume that the corresponding one-layer logit events $E_{\ell, N}$ satisfy

$$\sum_{\ell \in \mathcal{P}_N} \mathbb{P}(E_{\ell, N}^c) \rightarrow 0,$$

that their perturbation scales are summable,

$$\sum_{\ell \in \mathcal{P}_N} \frac{1}{|T|} \sum_{s \in T} a_{\phi, \ell}(N, s) \rightarrow 0,$$

and that the accumulated cross term is negligible,

$$\sum_{\ell \in \mathcal{P}_N} \langle S_\ell, R_\ell \rangle_F = o_{\mathbb{P}}(1).$$

Then the same telescoping conclusion holds with $\mathcal{P}$ replaced by $\mathcal{P}_N$.

Theorem 4.7. Assume Assumptions 4.1–4.2. Assume further that for all $s \in T$,

$$a_\phi(N, s) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Suppose:

1. for each N , $\alpha(t, N) \rightarrow \alpha^*(N)$ in probability as $t \rightarrow \infty$;
2. the relevant limit weight matrix admits an admissible decomposition $W^*(N) = S^*(N) + R^*(N)$ satisfying

$$\langle S^*(N), R^*(N) \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

For $\vartheta \in [0, 1]$, let $\alpha_{\vartheta}^*(N)$ denote the parameter vector obtained by replacing $W^*(N)$ with $S^*(N) + \vartheta R^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event

$$\liminf_{h \downarrow 0} \frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \geq -\epsilon_N$$

has probability tending to one. Define

$$\bar{a}_{\phi, N} := \frac{2}{|T|} \sum_{s \in T} a_{\phi}(N, s).$$

Then

$$\|R^*(N)\|_F^2 \leq \frac{\bar{a}_{\phi, N} + \epsilon_N}{2\lambda_{\text{reg}}} + o_{\mathbb{P}}(1).$$

In particular, if $\bar{a}_{\phi, N} + \epsilon_N \rightarrow 0$ in probability, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \mathbb{P}(\|R^*(N)\|_F \leq \epsilon) = 1.$$

If, in addition, there exists an admissible path decomposition

$$W(t, N) = S(t, N) + R(t, N)$$

such that for each fixed N ,

$$\|S(t, N) - S^*(N)\|_F + \|R(t, N) - R^*(N)\|_F \rightarrow 0 \quad \text{in probability as } t \rightarrow \infty,$$

and if for every fixed N and every $\epsilon > 0$,

$$\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0,$$

then

$$\lim_{N \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{P}(\|R(t, N)\|_F \leq \epsilon) = 1.$$

5 Main Theoretical Results: Deterministic Pruning Certificates

This section states deterministic data-path certificates based on $B_T(R)$. The mask-specific results concern fixed kept/removed supports and do not assume an additive Gaussian decomposition; random models only supply sufficient bounds for that budget. Proofs are collected in Appendix I.

5.1 Deterministic data-path certificates for mask pruning

Fix one target layer $A \in \mathbb{R}^{n \times m}$ inside a network and write the logits as

$$z_A(s) := \psi_2(A\psi_1(s)) \in \mathbb{R}^K, \quad s \in T,$$

where ψ_1 and ψ_2 denote the pre-layer and post-layer computations with all other parameters frozen. For a decomposition

$$A = S + R, \quad A_{\theta} := S + \theta R, \quad 0 \leq \theta \leq 1,$$

let L_s be any finite deterministic upper bound on the local ℓ_{∞} -Lipschitz constant of ψ_2 on the path

$$\{A_{\theta}\psi_1(s) : 0 \leq \theta \leq 1\}.$$

Define the data-path budget of the removed component by

$$B_T(R) := \frac{2}{|T|} \sum_{s \in T} L_s \|R\psi_1(s)\|_\infty. \quad (8)$$

If L_s is itself estimated from the realized network, the statements below are deterministic conditional on that realized upper bound.

Lemma 5.1 (Deterministic cross-entropy path bound). *For every decomposition $A = S + R$ and every $\theta \in [0, 1]$,*

$$|\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1)| \leq (1 - \theta)B_T(R),$$

where

$$\mathcal{L}_{\text{CE}}(A) := \frac{1}{|T|} \sum_{s \in T} \left[-z_A(s)_{C(s)} + \log \left(\sum_{j=1}^K e^{z_A(s)_j} \right) \right].$$

Corollary 5.2 (Deterministic margin stability). *Let*

$$\eta_s(\theta) := (1 - \theta)L_s \|R\psi_1(s)\|_\infty.$$

If

$$p_A(s) \in \arg \max_j z_A(s)_j, \quad \Delta_A^{\text{pred}}(s) := z_A(s)_{p_A(s)} - \max_{j \neq p_A(s)} z_A(s)_j,$$

then the predicted label at s can change between A_1 and A_θ only if

$$\Delta_{A_1}^{\text{pred}}(s) \leq 2\eta_s(\theta).$$

For the true-label margin

$$\Delta_A^{\text{true}}(s) := z_A(s)_{C(s)} - \max_{j \neq C(s)} z_A(s)_j,$$

one has

$$|\text{acc}_{A_\theta}(T) - \text{acc}_{A_1}(T)| \leq \frac{1}{|T|} \# \{s \in T : |\Delta_{A_1}^{\text{true}}(s)| \leq 2\eta_s(\theta)\}.$$

Here

$$\text{acc}_A(T) := \frac{1}{|T|} \# \{s \in T : \Delta_A^{\text{true}}(s) > 0\}.$$

Theorem 5.3 (Deterministic additive Frobenius certificate). *For any decomposition $A = S + R$, not necessarily a mask decomposition, define the one-layer Frobenius-regularized objective*

$$\mathcal{J}_2(A) := \mathcal{L}_{\text{CE}}(A) + \lambda_2 \|A\|_F^2 + \mathcal{C}, \quad \lambda_2 \geq 0,$$

where $\mathcal{C}$ contains all terms independent of A . Then for every $\theta \in [0, 1]$,

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - 2\lambda_2(1 - \theta)\langle S, R \rangle_F.$$

Consequently, if $|\langle S, R \rangle_F| \leq \eta$, then

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 + 2\lambda_2(1 - \theta)\eta.$$

In particular, full removal is certified to decrease the Frobenius-regularized objective whenever

$$B_T(R) + 2\lambda_2|\langle S, R \rangle_F| < \lambda_2\|R\|_F^2.$$

Theorem 5.4 (Deterministic elastic-net mask pruning certificate). *Assume $A = S + R$ is a mask decomposition, meaning*

$$S \odot R = 0.$$

For constants $\lambda_2, \lambda_1 \geq 0$, define the one-layer elastic-net objective by

$$\mathcal{J}(A) := \mathcal{L}_{\text{CE}}(A) + \lambda_2 \|A\|_F^2 + \lambda_1 \|A\|_1 + \mathcal{C},$$

where $\mathcal{C}$ contains all terms independent of A . Then for every $\theta \in [0, 1]$,

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - \lambda_1(1 - \theta)\|R\|_1.$$

For full pruning,

$$\mathcal{J}(S) \leq \mathcal{J}(A) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1 + B_T(R).$$

Consequently, the mask is certified to decrease the elastic-net objective whenever

$$B_T(R) < \lambda_2\|R\|_F^2 + \lambda_1\|R\|_1.$$

Theorem 5.5 (Deterministic prune–restore certificate). *Let the original layer decompose as*

$$A = S + Q + P, \quad S \odot Q = S \odot P = Q \odot P = 0.$$

Interpret P as the entries finally removed and Q as entries restored, or kept, from an over-pruned reservoir. Define

$$A_\theta := S + Q + \theta P, \quad 0 \leq \theta \leq 1.$$

For the elastic-net objective $\mathcal{J}$ of Theorem 5.4, one has

$$\mathcal{J}(A_\theta) - \mathcal{J}(A) \leq (1 - \theta)B_T(P) - \lambda_2(1 - \theta^2)\|P\|_F^2 - \lambda_1(1 - \theta)\|P\|_1.$$

For the final prune–restore layer $S + Q$,

$$\mathcal{J}(S + Q) \leq \mathcal{J}(A) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 + B_T(P).$$

Consequently, the final SER-style mask is certified to decrease the elastic-net objective whenever

$$B_T(P) < \lambda_2\|P\|_F^2 + \lambda_1\|P\|_1.$$

Corollary 5.6 (When restoration improves the certificate upper bound). *Under the hypotheses of Theorem 5.5, set $R := Q + P$. Comparing the all-pruned certificate for $A = S + R$ with the restore- Q certificate for $A = (S + Q) + P$, the restored upper bound is strictly smaller iff*

$$B_T(R) - B_T(P) > \lambda_2\|Q\|_F^2 + \lambda_1\|Q\|_1.$$

Coordinatewise weighted variants are stated in Appendix E.

Theorem 5.7 (One-sided stationarity controls removable masks). *Under the hypotheses of Theorem 5.4, assume the one-sided directional stationarity condition holds for some $\varepsilon \geq 0$:*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(A_{1-h}) - \mathcal{J}(A_1)}{h} \geq -\varepsilon.$$

Then

$$2\lambda_2\|R\|_F^2 + \lambda_1\|R\|_1 \leq B_T(R) + \varepsilon.$$

More generally, if

$$\mathcal{J}(A_\theta) \geq \mathcal{J}(A_1) - \varepsilon(1 - \theta) \quad \forall \theta \in [0, 1],$$

then for every $\theta < 1$,

$$\lambda_2(1 + \theta)\|R\|_F^2 + \lambda_1\|R\|_1 \leq B_T(R) + \varepsilon.$$

Theorem 5.8 (Deterministic multi-layer mask certificate). *Let $\mathcal{J}(\alpha)$ denote the full cross-entropy plus layerwise elastic-net objective. Consider an ordered finite sequence of mask-pruning operations on layers $j = 1, \dots, M$. At step j , write the current layer after steps $1, \dots, j-1$ as*

$$A_j = S_j + R_j, \quad S_j \odot R_j = 0,$$

and let $B_j(R_j)$ be its data-path budget in that intermediate network. If $\alpha^{(0)}$ and $\alpha^{(M)}$ denote the parameters before and after the M mask removals, then

$$\mathcal{J}(\alpha^{(M)}) \leq \mathcal{J}(\alpha^{(0)}) - \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1 + \sum_{j=1}^M B_j(R_j),$$

where $\lambda_{2,j}$ and $\lambda_{1,j}$ are the elastic-net weights applied to layer j .

The corresponding multi-layer margin-stability corollary is stated in Appendix E.

Theorem 5.9 (Multi-layer stationarity controls jointly removable masks). *Use the notation of Theorem 5.8. For $h \in [0, 1]$, let α_h be obtained by applying the same ordered pruning operations with each current layer $A_j = S_j + R_j$ replaced by $S_j + (1-h)R_j$. Assume that each $B_j(R_j)$ remains valid along this fractional path, uniformly for $h \in [0, 1]$, so the step- j cross-entropy increase is at most $hB_j(R_j)$. If the one-sided directional stationarity condition*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)})}{h} \geq -\varepsilon$$

holds for some $\varepsilon \geq 0$, then

$$2 \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 + \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1 \leq \sum_{j=1}^M B_j(R_j) + \varepsilon.$$

More generally, if

$$\mathcal{J}(\alpha_h) \geq \mathcal{J}(\alpha^{(0)}) - \varepsilon h \quad \forall h \in [0, 1],$$

then for every $h \in (0, 1]$,

$$\sum_{j=1}^M \lambda_{2,j} (2-h) \|R_j\|_F^2 + \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1 \leq \sum_{j=1}^M B_j(R_j) + \varepsilon.$$

Corollary 5.10 (Gaussian sufficient condition for the data-path budget). *Let $R \in \mathbb{R}^{n \times m}$ have independent entries $R_{ij} \sim N(0, g/n)$, independent of the collection $v_s := \psi_1(s)$, $s \in T$. Assume L_s are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of R . Then for every finite T and every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$B_T(R) \leq \frac{2}{|T|} \sum_{s \in T} L_s \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}}.$$

Corollary 5.11 (MP-edge form of the Gaussian data-path budget). *Under the hypotheses of Corollary 5.10, define the nominal rectangular MP singular-value edge of $R \in \mathbb{R}^{n \times m}$ by*

$$\sigma_{+,R} := \sqrt{g} \left(1 + \sqrt{\frac{m}{n}} \right).$$

Then, with probability at least $1 - \delta$,

$$B_T(R) \leq \frac{2\sigma_{+,R}}{1 + \sqrt{m/n}} \sqrt{\frac{2}{n} \log \frac{2n|T|}{\delta}} \frac{1}{|T|} \sum_{s \in T} L_s \|v_s\|_2.$$

Sub-Gaussian and simultaneous multi-layer budget corollaries are stated in Appendix E.

Acknowledgments

LB, HO, and YS acknowledge support from NASA via the AIST program (Kernel Flows: Emulating Complex Models for Massive Data Sets). This work started during LB’s sabbatical stay at Caltech hosted by H. Owahdi; LB and YS are grateful for the hospitality during their NASA-supported visit.

The work of LB was partially supported by NSF grant DMS-2005262 and NSF grant IMPRESS-U 2401227. TB and HO acknowledge support from the Air Force Office of Scientific Research under MURI award number FA9550-20-1-0358 (Machine Learning and Physics-Based Modeling and Simulation) and FOA-AFRL-AFOSR-2023-0004 (Mathematics of Digital Twins), and from the Department of Energy under award number DE-SC0023163 (SEA-CROGS: Scalable, Efficient, and Accelerated Causal Reasoning Operators, Graphs and Spikes for Earth and Embedded Systems). HO and TB also acknowledge support from the DoD Vannevar Bush Faculty Fellowship Program under award number ONR-N000142512035.

The work of YS was partially supported by the European Research Council under the European Union’s Horizon 2022 research and innovation program (grant agreement No. 101041711), the Simons Foundation, the Israel Science Foundation (grant number 2258/19), and the Israel Science Foundation (ISF Grant 4101/25).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, numerical interpretations, and bibliographic information were subsequently reviewed and edited by the author(s), who take full responsibility for the content of the manuscript.

Data availability

Code, plotting scripts, pruning recipes, and manuscript artifacts are available at https://github.com/yspennstate/RMT_based_pruning_in_deep_learning. The checkpoints, validation ledgers, sparse-backend timing evidence, and per-run files supporting the main ImageNet-1k tables are available at <https://drive.google.com/drive/folders/1mm990SHAHL1YdISHxirvMRdVQEAjpIxDd>. ImageNet-1k images are not redistributed; users must obtain the dataset under its own terms [12]. Appendices A and D and the companion methodology paper identify the source checkpoints, recipes, and evidence files for the reported rows.

References

- [1] Sangho An, Jinwoo Kim, Keonho Lee, Jingang Huh, Chanwoong Kwak, Yujin Lee, Moonsub Jin, and Jangho Kim. Sparse structure exploration and re-optimization for vision transformer. In *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence*, volume 286 of *Proceedings of Machine Learning Research*, pages 111–131. PMLR, 2025. URL <https://proceedings.mlr.press/v286/an25a.html>.
- [2] Florent Benaych-Georges and Raj Rao Nadakuditi. The singular values and vectors of low rank perturbations of large rectangular random matrices. *Journal of Multivariate Analysis*, 111:120–135, 2012.

- [3] Leonid Berlyand, Etienne Sandier, Yitzchak Shmalo, and Lei Zhang. Enhancing accuracy in deep learning using random matrix theory. *Journal of Machine Learning*, 3(4):347–412, 2024. doi: 10.4208/jml.231220.
- [4] Leonid Berlyand, Theo Bourdais, Houman Owhadi, and Yitzchak Shmalo. Pruning deep neural networks via a combination of the marchenko–pastur distribution and regularization, 2025. URL <https://arxiv.org/abs/2503.01922>.
- [5] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your ViT but faster. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=JroZRw7Eu>.
- [6] Chenghao Cai, Dengfeng Ke, Yanyan Xu, and Kaile Su. Fast learning of deep neural networks via singular value decomposition. In *Pacific Rim International Conference on Artificial Intelligence*, pages 820–826, 2014.
- [7] Tianlong Chen, Yu Cheng, Zhe Gan, Lu Yuan, Lei Zhang, and Zhangyang Wang. Chasing sparsity in vision transformers: An end-to-end exploration. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [8] Hongrong Cheng, Miao Zhang, and Javen Qinfeng Shi. A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10558–10578, 2024. doi: 10.1109/TPAMI.2024.3447085.
- [9] Anna Choromanska, Mikael Henaff, Michael Mathieu, Gérard Ben Arous, and Yann LeCun. The loss surfaces of multilayer networks. In *Artificial Intelligence and Statistics*, pages 192–204, 2015.
- [10] Romain Couillet and Mérouane Debbah. *Random Matrix Methods for Wireless Communications*. Cambridge University Press, 2011.
- [11] Yann N. Dauphin, Razvan Pascanu, Caglar Gulcehre, Kyunghyun Cho, Surya Ganguli, and Yoshua Bengio. Identifying and attacking the saddle point problem in high-dimensional non-convex optimization. In *Advances in Neural Information Processing Systems*, volume 27, 2014.
- [12] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.
- [13] Xiaohan Ding, Tianxiang Hao, Jianchao Tan, Ji Liu, Jungong Han, Yuchen Guo, and Guiguang Ding. ResRep: Lossless CNN pruning via decoupling remembering and forgetting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4510–4520, 2021.
- [14] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.
- [15] Gongfan Fang, Xinyin Ma, Mingli Song, Michael Bi Mi, and Xinchao Wang. DepGraph: Towards any structural pruning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16091–16101, 2023.
- [16] Gongfan Fang, Xinyin Ma, Michael Bi Mi, and Xinchao Wang. Isomorphic pruning for vision models. In *European Conference on Computer Vision*. Springer, 2024.

- [17] Elias Frantar and Dan Alistarh. SparseGPT: Massive language models can be accurately pruned in one-shot. In *ICML*, 2023.
- [18] Jungang Ge, Ying-Chang Liang, Zhidong Bai, and Guangming Pan. Large-dimensional random matrix theory and its applications in deep learning and wireless communications. *Random Matrices: Theory and Applications*, 10(4):2230001, 2021.
- [19] Saurabh Goyal, Anamitra Roy Choudhury, Saurabh Raje, Venkatesan Chakaravarthy, Yogish Sabharwal, and Ashish Verma. PoWER-BERT: Accelerating BERT inference via progressive word-vector elimination. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 3690–3699. PMLR, 2020.
- [20] Yang He, Guoliang Kang, Xuanyi Dong, Yanwei Fu, and Yi Yang. Soft filter pruning for accelerating deep convolutional neural networks. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*, pages 2234–2240, 2018. doi: 10.24963/ijcai.2018/309.
- [21] Yang He, Ping Liu, Ziwei Wang, Zhilan Hu, and Yi Yang. Filter pruning via geometric median for deep convolutional neural networks acceleration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4340–4349, 2019.
- [22] Geoffrey Hinton, Li Deng, Dong Yu, George E. Dahl, Abdel-rahman Mohamed, Navdeep Jaitly, Andrew Senior, Vincent Vanhoucke, Patrick Nguyen, Tara N. Sainath, et al. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal Processing Magazine*, 29(6):82–97, 2012.
- [23] Ning-Chi Huang, Chi-Chih Chang, Wei-Cheng Lin, Endri Taka, Diana Marculescu, and Kai-Chiang Wu. ELSA: Exploiting layer-wise N:M sparsity for vision transformer acceleration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 8006–8015, June 2024.
- [24] Ashish Khetan and Zohar Karnin. Prunenet: Channel pruning via global importance. In *1st Workshop on Neural Architecture Search at ICLR 2020*, 2020. URL <https://www.amazon.science/publications/prunenet-channel-pruning-via-global-importance>.
- [25] Zhenglun Kong, Peiyan Dong, Xiaolong Ma, Xin Meng, Wei Niu, Mengshu Sun, Xuan Shen, Geng Yuan, Bin Ren, Hao Tang, Minghai Qin, and Yanzhi Wang. Spvit: Enabling faster vision transformers via latency-aware soft token pruning. In *Computer Vision – ECCV 2022*, pages 620–640. Springer, 2022. doi: 10.1007/978-3-031-20083-0_37.
- [26] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6):84–90, 2017.
- [27] Anders Krogh and John A. Hertz. A simple weight decay can improve generalization. In *Advances in Neural Information Processing Systems*, volume 4, pages 950–957, 1991. URL <https://papers.nips.cc/paper/1991/hash/8eefcfd5990e441f0fb6f3fad709e21-Abstract.html>.
- [28] Denis Kuznedelev, Eldar Kurtic, Elias Frantar, and Dan Alistarh. oViT: An accurate second-order pruning framework for vision transformers. OpenReview, 2023. URL https://openreview.net/forum?id=zYWtq_HUCoi.
- [29] Yann LeCun, Bernhard E. Boser, John S. Denker, Donnie Henderson, R. E. Howard, Wayne E. Hubbard, and Lawrence D. Jackel. Handwritten digit recognition with a back-propagation network. In *Advances in Neural Information Processing Systems*, volume 2, 1989. URL https://papers.nips.cc/paper_files/paper/1989/hash/53c3bce66e43be4f209556518c2fcb54-Abstract.html.

- [30] Youwei Liang, Chongjian Ge, Zhan Tong, Yibing Song, Jue Wang, and Pengtao Xie. Evit: Expediting vision transformers via token reorganizations. In *International Conference on Learning Representations*, 2022.
- [31] Mingbao Lin, Rongrong Ji, Yan Wang, Yichen Zhang, Baochang Zhang, Yonghong Tian, and Ling Shao. HRank: Filter pruning using high-rank feature map. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1529–1538, 2020.
- [32] Haiquan Lu, Yefan Zhou, Shiwei Liu, Zhangyang Wang, Michael W. Mahoney, and Yaoqing Yang. AlphaPruning: Using heavy-tailed self regularization theory for improved layer-wise pruning of large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0289.
- [33] Michael W. Mahoney and Charles H. Martin. Traditional and heavy-tailed self-regularization in neural network models. In *International Conference on Machine Learning*, pages 4284–4293, 2019.
- [34] Vladimir A. Marchenko and Leonid A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Matematicheskii Sbornik*, 114(4):507–536, 1967.
- [35] Charles H. Martin and Michael W. Mahoney. Heavy-tailed universality predicts trends in test accuracies for very large pre-trained deep neural networks. In *Proceedings of the 2020 SIAM International Conference on Data Mining*, pages 505–513, 2020.
- [36] Charles H. Martin and Michael W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *Journal of Machine Learning Research*, 22(165):1–73, 2021.
- [37] Charles H. Martin, Tongsu Peng, and Michael W. Mahoney. Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data. *Nature Communications*, 12(4122), 2021.
- [38] Xuran Meng and Jianfeng Yao. Impact of classification difficulty on the weight matrices spectra in deep learning and application to early-stopping. *Journal of Machine Learning Research*, 24: 1–40, 2023.
- [39] Asit Mishra, Jorge Albericio Latorre, Jeff Pool, Darko Stosic, Dusan Stosic, Ganesh Venkatesh, Chong Yu, and Paulius Micikevicius. Accelerating sparse deep neural networks. *arXiv preprint arXiv:2104.08378*, 2021. doi: 10.48550/arXiv.2104.08378.
- [40] Raj Rao Nadakuditi. Optshrink: An algorithm for improved low-rank signal matrix denoising by optimal, data-driven singular value shrinkage. *IEEE Transactions on Information Theory*, 60(5): 3002–3018, 2014. doi: 10.1109/TIT.2014.2311661.
- [41] Thiziri Nait Saada and Jared Tanner. On the initialisation of wide low-rank feedforward neural networks. *arXiv preprint arXiv:2301.13710*, 2023.
- [42] Zizheng Pan, Bohan Zhuang, Jing Liu, Haoyu He, and Jianfei Cai. Scalable vision transformers with hierarchical pooling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 377–386, 2021.
- [43] Jiyoung Park, Ian Pelakh, and Stephan Wojtowytsch. Minimum norm interpolation by perceptrons: Explicit regularization and implicit bias. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

- [44] Leonid Pastur. On random matrices arising in deep neural networks. gaussian case. *Pure and Applied Functional Analysis*, 5(6):1395–1424, 2020.
- [45] Leonid Pastur and Victor Slavin. On random matrices arising in deep neural networks: General iid case. *Random Matrices: Theory and Applications*, 12(1):2250046, 2023.
- [46] Lutz Prechelt. Early stopping—but when? In *Neural Networks: Tricks of the Trade*, pages 53–67. Springer, 2012.
- [47] Yongming Rao, Wenliang Zhao, Benlin Liu, Jiwen Lu, Jie Zhou, and Cho-Jui Hsieh. Dynamicvit: Efficient vision transformers with dynamic token sparsification. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [48] Vadim I. Serdobolskii. *Multivariate Statistical Analysis: A High-Dimensional Approach*. Springer, 2000.
- [49] Yitzchak Shmalo, Jonathan Jenkins, and Oleksii Krupchytskyi. Deep learning weight pruning with rmt-svd: Increasing accuracy and reducing overfitting. *arXiv preprint arXiv:2303.08986*, 2023.
- [50] Zhuoran Song, Yihong Xu, Zhezhi He, Li Jiang, Naifeng Jing, and Xiaoyao Liang. Cp-vit: Cascade vision transformer pruning via progressive sparsity prediction. *arXiv preprint arXiv:2203.04570*, 2022.
- [51] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958, 2014.
- [52] Max Staats, Matthias Thamm, and Bernd Rosenow. Boundary between noise and information applied to filtering neural network weight matrices. *Physical Review E*, 108(2):L022302, 2023. doi: 10.1103/PhysRevE.108.L022302.
- [53] G. W. Stewart and Ji-guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- [54] Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. A simple and effective pruning approach for large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=PxoFut3dWW>.
- [55] Xinglong Sun, Barath Lakshmanan, Maying Shen, Shiyi Lan, Jingde Chen, and Jose M. Alvarez. Mdp: Multidimensional vision model pruning with latency constraint. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20113–20123, 2025.
- [56] Ilya Sutskever, Oriol Vinyals, and Quoc V. Le. Sequence to sequence learning with neural networks. In *Advances in Neural Information Processing Systems*, volume 27, pages 3104–3112, 2014.
- [57] Kai Sheng Tai, Taipeng Tian, and Ser-Nam Lim. Spartan: Differentiable sparsity via regularized transportation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [58] Matthias Thamm, Max Staats, and Bernd Rosenow. Random matrix analysis of deep neural network weight matrices. *Physical Review E*, 106(5):054124, 2022.
- [59] Sunil Vadera and Salem Ameen. Methods for pruning deep neural networks. *IEEE Access*, 10: 63280–63300, 2022.

- [60] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [61] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.
- [62] Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12(1):99–111, 1972.
- [63] Xuanzhe Xiao, Zeng Li, Chuanlong Xie, and Fengwei Zhou. Heavy-tailed regularization of weight matrices in deep neural networks. In *Artificial Neural Networks and Machine Learning – ICANN 2023*. Springer Nature Switzerland, 2023. doi: 10.1007/978-3-031-44204-9_20.
- [64] Anhao Xing, Pengyuan Zhang, Jieli Pan, and Yonghong Yan. Svd-based dnn pruning and retraining. *Journal of Tsinghua University (Science and Technology)*, 56(7):772–776, 2016.
- [65] Yuhui Xu, Yuxi Li, Shuai Zhang, Wei Wen, Botao Wang, Wenrui Dai, Yingyong Qi, Yiran Chen, Weiyao Lin, and Hongkai Xiong. Trained rank pruning for efficient deep neural networks. In *Fifth Workshop on Energy Efficient Machine Learning and Cognitive Computing–NeurIPS Edition*, pages 14–17, 2019.
- [66] Jian Xue, Jinyu Li, and Yifan Gong. Restructuring of deep neural network acoustic models with singular value decomposition. In *INTERSPEECH*, pages 2365–2369, 2013.
- [67] Huanrui Yang, Minxue Tang, Wei Wen, Feng Yan, Daniel Hu, Ang Li, Hai Li, and Yiran Chen. Learning low-rank deep neural networks via singular vector orthogonality regularization and singular value sparsification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 2899–2908, 2020. doi: 10.1109/CVPRW50498.2020.00347.
- [68] Huanrui Yang, Hongxu Yin, Maying Shen, Pavlo Molchanov, Hai Li, and Jan Kautz. Global vision transformer pruning with hessian-aware saliency. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18547–18557, 2023.
- [69] Hongxu Yin, Arash Vahdat, Jose M. Alvarez, Arun Mallya, Jan Kautz, and Pavlo Molchanov. A-vit: Adaptive tokens for efficient vision transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10809–10818, 2022.
- [70] Chong Yu, Tao Chen, Zhongxue Gan, and Jiayuan Fan. Boost vision transformer with GPU-friendly sparsity and quantization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22658–22668, June 2023. doi: 10.1109/CVPR52729.2023.02170.
- [71] Lu Yu and Wei Xiang. X-pruner: explainable pruning for vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24355–24363, 2023.
- [72] Hanxiao Zhang, Yifan Zhou, and Guo-Hua Wang. Dense vision transformer compression with few samples. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15825–15834, 2024.
- [73] Jinnian Zhang, Houwen Peng, Kan Wu, Mengchen Liu, Bin Xiao, Jianlong Fu, and Lu Yuan. MiniViT: Compressing vision transformers with weight multiplexing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12145–12154, 2022.

- [74] Kang Zhao, Tao Yuan, Han Bao, Zhenfeng Su, Chang Gao, Zhaofeng Sun, Zichen Liang, Liping Jing, and Jianfei Chen. Beyond 2:4: Exploring V:N:M sparsity for efficient transformer inference on GPUs, 2024. URL <https://arxiv.org/abs/2410.16135>.
- [75] Chuanyang Zheng, Zheyang Li, Kai Zhang, Zhi Yang, Wenming Tan, Jun Xiao, Ye Ren, and Shiliang Pu. Savit: Structure-aware vision transformer pruning via collaborative optimization. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [76] Mingjian Zhu, Yehui Tang, and Kai Han. Vision transformer pruning. *arXiv preprint arXiv:2104.08500*, 2021. Accepted by the KDD 2021 Workshop on Model Mining.

A CAST/CAST-2E Recipe Table

Table 8 records the compact recipe summary for the main structured-projection rows.

Table 8: Hyperparameters for representative CAST/CAST-2E rows of Table 3. The table lists the source checkpoint for each run and the post-projection distillation schedule. The CAST projection (Algorithm 2) is gradient-free.

	ViT-B/16 6:12	ViT-B/16 2:4+ToMe	ViT-L/16 2:4+ToMe	ResNet50d 8:16
Source ckpt	SER $s = 0.35$	SER $s = 0.35$	SER $s = 0.35$	dense (no SER)
Calibration set $ C $	256	256	256	64
Permute align	yes	no	no	yes
Free restoration	yes	yes	yes	yes
α_{ser}	0.5	0.1	0.1	0.0
ToMe r	—	8	8	—
<i>Stage 2</i>				
<i>distillation:</i>				
Optimiser	AdamW	AdamW	AdamW	SGD+momentum 0.9
Base LR	1×10^{-4}	1×10^{-5}	1×10^{-5}	1×10^{-3}
Weight decay	0.05	0.01	0.01	1×10^{-4}
LR schedule	cosine + warmup	cosine + warmup	cosine + warmup	cosine + warmup
Warmup steps	1000	1000	1000	500
Train batch	256	32	8	256
Val batch	128	64	16	128
Label smoothing	0.1	0.1	0.1	0.1
ϵ				
Distill (α, τ)	(0.5, 2.0)	(0.5, 2.0)	(0.5, 2.0)	(0.5, 2.0)
Epochs	3	3	3	3
Mask re-zero	every step	every step	every step	every step
Post-FT top-1	Reported in Table 3			

B Prior-Pruning Context

Classical applications motivating DNN compression include handwriting recognition [29], image classification [26], speech recognition [22], and natural language processing [56]. Standard overfitting countermeasures include dropout [51], early stopping [46], and weight decay [27, 43]. DNN training searches a high-dimensional nonconvex loss landscape with local minima, saddle points, and flat regions [11, 9]. RMT has also been used for stopping criteria and other deep-learning diagnostics. Vadera and Ameen group pruning methods into magnitude pruning, redundancy/clustering, and sensitivity analysis [59]; Cheng et al. organize the broader field by speedup, pruning timing, pruning criterion,

and interaction with other compression techniques [8]; PruneNet is a representative channel-pruning method [24]. Prior SVD-based compression uses low-rank restructuring or singular-value truncation, with rank choices ranging from fixed/data-driven singular-value rules to energy or validation heuristics [66, 6, 64, 67, 65]. Earlier work by our group reported MP-based compression on small fully connected DNNs and MNIST-scale spectral pruning [3, 4, 49].

ViT compression spans several axes. Token-level methods reduce sequence length through dynamic token pruning [47, 30, 69, 25] or token merging [5] without changing dense projection weights, while block/head/low-rank and multi-granularity methods alter other structural axes. oViT studies second-order ViT weight sparsification [28]; block/head/module-level structured pruning is studied in [75, 68, 16, 71, 72]; MiniViT studies weight multiplexing [73]; SViTE and Spartan study sparse training from scratch [7, 57]; MDP studies multi-granularity latency-aware pruning [55]. Recent unstructured pruning at scale uses Hessian-based one-shot signals [17], magnitude-weighted-by-activation signals [54], and heavy-tailed self-regularization signals [32, 35].

C Theory–Algorithm–Result Mapping

Table 9: Algorithms, associated RMT/certificate statements, inputs, and reported empirical rows.

Algorithm	Associated RMT / certificate statement	Inputs / what it computes	Reported row
Spectral Edge Reallocation (SER, Algorithm 1)	Corollary 5.11; Theorem 5.5.	BEMA MP edges $\sigma_+(\ell) \rightarrow$ standardized layer signal $z_\ell \rightarrow$ over-pruned reservoir $\rightarrow$ RMT-ranked reinsertion $\rightarrow$ frozen-mask short FT.	Hybrid Mag-SER multi-arch row, Table 2.
Hybrid SER	Magnitude-SER plus the empirical sweep result that magnitude pruning remains competitive at $s \leq 0.20$, so RMT modulation is activated for $s \geq 0.25$.	Exact magnitude through $s = 0.20$, then SER continuation with one epoch per pruning cycle.	ViT-B/16 and 17-checkpoint rows in Tables 1 and 2.
CAST ($k:n$ projection) (Algorithm 2)	Lemma 5.1; Theorem 5.5; Corollary 5.6.	SER source mask M^{SER} , dense pre-trained W^{dense} , calibration set $\mathcal{C}$, within-group covariance C_g , per-row argmin over $\binom{n}{k}$ patterns, and optional free restoration.	CAST rows in Table 3.
Cin permutation alignment	Empirical importance-balanced input-channel ordering before the per-row argmin; consistent with the layerwise sum form of Theorem 5.8.	Score $s_j = \sum_i W_{i,j}^{\text{dense}} h_j $, quartile-interleave channels, solve in permuted basis, invert permutation.	ResNet and ConvNeXtV2 dense+perm rows.
2:4 deployable backend audit	The certificate framework requires only a fixed mask; NVIDIA Sparse Tensor Cores execute the fixed 2:4 structure [39].	Load checkpoint, convert to Linear layers with <code>torch.sparse.to_sparse_semi_structured</code> ; ResNet audit lowers Conv2d to <code>im2col+Linear</code> .	el- Table 4.

D Comparison Provenance and Scope Notes

Training budgets already shown in Table 6 are not repeated here. The AlphaPruning-style row is an author reimplement under this paper’s data, checkpoint, layer set, and evaluation protocol because [32] does not report that exact ViT-B/16 ImageNet cell. Zero-fine-tuning author-run one-shot baselines and external rows without a reported training/fine-tuning budget are omitted from the main comparison table; the omitted values are SparseGPT-style 50% unstructured ViT-B/16 no-FT at $\sim 81.1\%$ (~ -4.0 pp), Wanda-style at $\sim 81.3\%$ (~ -3.8 pp), and LPViT’s DeiT-Base 50% FLOP block-sparse row at 80.81% (-0.99 pp) with no FT budget reported in the cited table. Spartan reports

its positive Δ against its own 80.06% ViT-B/16 dense model at the tabled training budget; against the DeiT-B baseline used elsewhere in the paper, the gap would be -0.62 pp.

CP-ViT’s DeiT-B row and its VTP/PoWER-BERT/HVT reproductions are the CP-ViT-reported entries in Table 6; its dense DeiT-B baseline is 81.82%. ELSA’s main uniform row is the one listed in Table 6; its Table 1 also reports layer-wise $N:4$ entries at 7.0G/81.6% and 6.0G/81.4%. These rows evaluate structured FLOP-reduction methods rather than $k:n$ weight sparsity, so they are DeiT-family context rather than a matched weight-sparsity benchmark. For this paper’s rows, the prunable-layer list is fixed across the multi-architecture sweep; masks are constructed without ImageNet validation access; validation is used only to report completed checkpoints and in documented global-protocol sweeps. Results are reported against the corresponding `timm` dense checkpoint.

Table 4 reports the selected best-observed A40 batch-sweep endpoints; batch 256 was also swept but not selected. The older A100 no-ToMe ViT-B endpoint remains separate: $463.6 \rightarrow 1254.1$ images/s, $2.705\times$, and should not be pooled with the ToMe-held-fixed A40 audits. ToMe-only AugReg references from [5] are 83.94% at $r = 8$ and 82.86% at $r = 12$.

Extending the ResNet endpoint caveat in Table 4, the `im2col+2:4` rows do not establish TensorRT or cuDNN Conv2d acceleration. The 1×1 -only controls remain below native cuDNN Conv2d throughput; see the methodology paper, quick-reference key `subsec:6to24`. A ResNet152d 8:16 dense+perm row is outside the reported fixed-calibration sweep: the wider-pattern certificate projection exceeded a single 80 GB A100 at the default calibration batch.

The wider 8:16 group expands the per-row search space from $\binom{4}{2} = 6$ to $\binom{16}{8} = 12,870$ patterns at the same nominal sparsity, and the dense source removes the SER-mask Hamming bias against weights previously discarded by unstructured pruning. The ViT-L/16 8:16 row has released evidence and checkpoint-status metadata in `data/release_manifest.json`.

The public artifact bundle is indexed by `data/release_manifest.json`; the companion methodology paper gives protocol-level provenance and ledger details. CAST row estimates are primarily single-seed; the available replicate summaries provide the repeatability check for the reported CAST setting.

E Additional Deterministic Certificate Corollaries

Corollary E.1 (Weighted elastic-net mask certificate). *Under the hypotheses of Theorem 5.4, let $\Gamma_2 = (\gamma_{2,ij})$ and $\Gamma_1 = (\gamma_{1,ij})$ be nonnegative coordinatewise weights, and define*

$$\mathcal{J}_\Gamma(A) := \mathcal{L}_{\text{CE}}(A) + \sum_{i,j} \gamma_{2,ij} A_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |A_{ij}| + \mathcal{C}.$$

Then, for every $\theta \in [0, 1]$,

$$\mathcal{J}_\Gamma(A_\theta) - \mathcal{J}_\Gamma(A_1) \leq (1 - \theta)B_T(R) - (1 - \theta^2) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - (1 - \theta) \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

In particular, full pruning is certified to decrease the weighted elastic-net objective whenever

$$B_T(R) < \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Corollary E.2 (Weighted stationarity controls removable masks). *Under the hypotheses of Corollary E.1, assume the one-sided directional stationarity condition*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \geq -\varepsilon$$

holds for some $\varepsilon \geq 0$. Then

$$2 \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}| \leq B_T(R) + \varepsilon.$$

More generally, if

$$\mathcal{J}_\Gamma(A_\theta) \geq \mathcal{J}_\Gamma(A_1) - \varepsilon(1 - \theta) \quad \forall \theta \in [0, 1],$$

then for every $\theta < 1$,

$$(1 + \theta) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}| \leq B_T(R) + \varepsilon.$$

Corollary E.3 (Deterministic multi-layer margin stability). *Use the ordered pruning sequence and notation of Theorem 5.8. Let $z^{(j)}(s)$ be the logit vector after the j -th mask operation, with $z^{(0)}$ denoting the original network and $z^{(M)}$ the final pruned network. Suppose that for each $s \in T$ and each step j there is a deterministic bound $\eta_{j,s} \geq 0$ such that*

$$\|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \leq \eta_{j,s}.$$

Define $\eta_s^{\text{tot}} := \sum_{j=1}^M \eta_{j,s}$. Then

$$|\mathcal{L}_{\text{CE}}(\alpha^{(M)}) - \mathcal{L}_{\text{CE}}(\alpha^{(0)})| \leq \frac{2}{|T|} \sum_{s \in T} \eta_s^{\text{tot}}.$$

Moreover, if $p_0(s) \in \arg \max_j z_j^{(0)}(s)$ and

$$\Delta_0^{\text{pred}}(s) := z_{p_0(s)}^{(0)}(s) - \max_{j \neq p_0(s)} z_j^{(0)}(s),$$

then the predicted label at s can change between the original and final networks only if

$$\Delta_0^{\text{pred}}(s) \leq 2\eta_s^{\text{tot}}.$$

For the true-label margin

$$\Delta_0^{\text{true}}(s) := z_{C(s)}^{(0)}(s) - \max_{j \neq C(s)} z_j^{(0)}(s),$$

the accuracy change obeys

$$|\text{acc}_{\alpha^{(M)}}(T) - \text{acc}_{\alpha^{(0)}}(T)| \leq \frac{1}{|T|} \#\{s \in T : |\Delta_0^{\text{true}}(s)| \leq 2\eta_s^{\text{tot}}\}.$$

Corollary E.4 (Sub-Gaussian sufficient condition for the data-path budget). *Let $v_s = \psi_1(s)$, and assume L_s are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of R . Let the rows $r_1, \dots, r_n$ of R be independent and centered conditionally on those variables, and suppose that for some $c, g > 0$,*

$$\mathbb{P}(|\langle r_i, v \rangle| > t \mid v) \leq 2 \exp\left(-\frac{cnt^2}{g\|v\|_2^2}\right)$$

for all i, v , and $t > 0$. Then, with probability at least $1 - \delta$,

$$B_T(R) \leq \frac{2}{|T|} \sum_{s \in T} L_s \|v_s\|_2 \sqrt{\frac{g}{cn} \log \frac{2n|T|}{\delta}}.$$

Corollary E.5 (Simultaneous multi-layer Gaussian data-path budget). *Consider the ordered finite pruning sequence in Theorem 5.8. At step j , let the removed component be modeled by $R_j \in \mathbb{R}^{n_j \times m_j}$ with independent $N(0, g_j/n_j)$ entries, conditionally independent of the intermediate pre-layer vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. Let $\delta_1, \dots, \delta_M > 0$ satisfy $\sum_{j=1}^M \delta_j \leq \delta$. Then, with probability at least $1 - \delta$, all layerwise budgets satisfy*

$$B_j(R_j) \leq \frac{2}{|T|} \sum_{s \in T} L_{j,s} \|v_{j,s}\|_2 \sqrt{\frac{2g_j}{n_j} \log \frac{2n_j|T|}{\delta_j}} \quad (1 \leq j \leq M).$$

Consequently, on the same event, Theorem 5.8 holds with $\sum_j B_j(R_j)$ replaced by the sum of the displayed right-hand sides.

Corollary E.6 (Simultaneous multi-layer sub-Gaussian data-path budget). *Use the ordered pruning sequence and notation of Corollary E.5, but at each step j replace the Gaussian model by the row-concentration hypothesis of Corollary E.4 with (R, n, c, g, v_s, L_s) replaced by $(R_j, n_j, c_j, g_j, v_{j,s}, L_{j,s})$. Let $\delta_1, \dots, \delta_M > 0$ satisfy $\sum_{j=1}^M \delta_j \leq \delta$. Then, with probability at least $1 - \delta$,*

$$B_j(R_j) \leq \frac{2}{|T|} \sum_{s \in T} L_{j,s} \|v_{j,s}\|_2 \sqrt{\frac{g_j}{c_j n_j} \log \frac{2n_j|T|}{\delta_j}} \quad (1 \leq j \leq M).$$

Consequently, the same deterministic multi-layer certificate applies with the displayed sub-Gaussian bounds.

F Notes on the Abstract Additive Framework

F.1 Relocated abstract additive corollaries

Corollary F.1 (Growing-set abstract loss reduction). *Let T_N be a finite labeled set that may depend on N . Let L_{reg, T_N} denote cross-entropy averaged over T_N , and let L_{T_N} denote the corresponding regularized objective with the same Frobenius penalty as in (7). Assume Assumptions 4.1–4.2,*

$$\langle S, R \rangle_F = o_{\mathbb{P}}(1), \quad |T_N| d_2(N) \rightarrow 0,$$

and

$$\frac{1}{|T_N|} \sum_{s \in T_N} a_\phi(N, s) \rightarrow 0$$

in probability. Then

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\text{reg}} \|R\|_F^2 + o_{\mathbb{P}}(1).$$

Corollary F.2. *Under the hypotheses of Theorem 4.5, for any fixed $\vartheta \in [0, 1]$, define $W(\vartheta) = S + \vartheta R$. Then*

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2) \|R\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0, 1]$.

Corollary F.3 (Finite- N scaled loss bound along the denoising path). *Assume the hypotheses of Theorem 4.5, and define $W(\vartheta) = S + \vartheta R$ for $\vartheta \in [0, 1]$. Then for every $\eta > 0$ there exists an event $E_{N,\eta}$ with*

$$\mathbb{P}(E_{N,\eta}) \geq 1 - |T| d_2(N) - \mathbb{P}(|\langle S, R \rangle_F| > \eta)$$

such that on $E_{N,\eta}$,

$$L(\alpha_{W(\vartheta)}) \leq L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R\|_F^2 + (1 - \vartheta)\frac{2}{|T|} \sum_{s \in T} a_\phi(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta$$

for every $\vartheta \in [0, 1]$.

Corollary F.4 (Eventual supercriticality of persistent spike clusters in a BGN-type deformation regime). *Under the hypotheses of Theorem 4.7, assume additionally that for each N , $W^*(N) = S^*(N) + R^*(N)$ lies in a finite-rank deformation regime satisfying all hypotheses of Theorem J.4 for the spikes considered below. Assume the regime has deterministic bulk edge $b_+(N)$, critical threshold $\theta_c(N)$, strictly decreasing outlier transform*

$$D_N : (b_+(N), \infty) \rightarrow (0, D_N(b_+(N)^+)),$$

local Lipschitz/invertibility intervals around the outliers, and concentration of the relevant sample singular values in those intervals. Assume $\theta_c(N) \rightarrow 0$, $b_+(N) \rightarrow 0$, and that for some fixed $k \geq 1$ there exist constants $\bar{\sigma}_1 > \dots > \bar{\sigma}_k > 0$ such that $\sigma_i^(N) \rightarrow \bar{\sigma}_i$ in probability for each $1 \leq i \leq k$. Then:*

1. $\mathbb{P}(\sigma_i^*(N) > \theta_c(N)) \rightarrow 1$ for each $1 \leq i \leq k$;
2. *these k spikes are eventually supercritical, so the corresponding singular values of $W^*(N)$ form outlier clusters above $b_+(N)$. If the i -th spike is separated from the other nonzero singular values of $S^*(N)$ by a gap $\Delta_i(N) > 0$ with $\|R^*(N)\|_{\text{op}}/\Delta_i(N) \rightarrow 0$ in probability, then the associated rank-one left and right spectral projectors converge in operator norm to the corresponding projectors of $S^*(N)$;*
3. *defining $\hat{\sigma}_i^{(D)}(N) := D_N(s_i(W^*(N)))^{-1/2}$ on the event $s_i(W^*(N)) > b_+(N)$, whose probability tends to one, and arbitrarily off that event, one has $\hat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \rightarrow 0$ in probability.*

Assumptions 4.1–4.2 are the abstract data-path version of the Gaussian conditions in Appendix G; Assumption G.3 gives the iid-Gaussian sufficient condition, while the optional ESD condition is used only for spectral conclusions.

Example F.5. The square iid-Gaussian case with variance $1/N$ gives Assumption 4.2 with $J(v) = \|v\|_2$, $d_1(N) = 2\sqrt{2\log(2N)/N}$, and $d_2(N) = 1/N$; see Corollary 5.10. The conditional formulation also accommodates sub-Gaussian rows, orthogonally invariant perturbations, or moderate training-induced dependence when the concentration estimate survives.

Example F.6 (Sub-Gaussian rows). The row-concentration variant in Corollary E.4 gives the corresponding Assumption 4.2 sufficient condition.

Assumption 4.3 is used only for spectral conclusions. Loss reduction uses the data-path smallness $a_\phi(N, s) \rightarrow 0$ and the cross-term condition; stationarity-collapse results additionally use the stated one-sided stationarity hypothesis.

Remark F.7 (Regularisation scaling). Throughout the loss-reduction statements, the elastic-net coefficients λ_1, λ_2 are treated as specified constants. An iid-Gaussian admissible component $R \in \mathbb{R}^{N \times N}$ with variance g/N has $\|R\|_F^2 = \Theta(gN)$ in expectation, so the loss-reduction term $\lambda_2\|R\|_F^2$ scales as $\Theta(\lambda_2 gN)$. The certificate inequality of Theorem 5.4 holds for any specified coefficient scaling, including coefficients normalized by parameter count or layer width. The data-path budget $B_T(R)$ is independent of λ_1, λ_2 , while the elastic-net reward is not; only the numerical certificate margin changes with regularizer scale.

Remark F.8 (Beyond pure Frobenius regularization). The quadratic penalty in (7) supplies a coercive lower bound of order $\|R\|_F^2$. For a differentiable μ -strongly convex regularizer Ω ,

$$\Omega(S + R) - \Omega(S) = \langle \nabla \Omega(S), R \rangle + D_\Omega(S + R, S), \quad D_\Omega(S + R, S) \geq \frac{\mu}{2} \|R\|_F^2.$$

The extension is valid when the first-order term is controlled or negligible. In the Frobenius case this term is $2\langle S, R \rangle_F$. For elastic net, choose $\xi \in \partial\|S\|_1$ and assume $\langle \mu_1 \xi + 2\mu_2 S, R \rangle = o_{\mathbb{P}}(1)$; the quadratic part supplies the strong convexity.

The stationarity theorem assumes only one-sided approximate stationarity at $\vartheta = 1$; exact local minimality is sufficient but stronger. Without the no-atom condition $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$, the same proof gives limsup/liminf threshold bounds rather than exact convergence. Corollary F.4 is the corresponding abstract BBP persistence statement.

Dependency map: deterministic mask certificates use the support condition

$$S = M \odot A, \quad R = (1 - M) \odot A,$$

so $S \odot R = 0$. Abstract additive loss statements replace that support condition by the relevant cross-term control, such as $\langle S, R \rangle_F = o_{\mathbb{P}}(1)$. Random budget corollaries provide sufficient conditions for $B_T(R)$.

Remark F.9 (Local Lipschitz constants L_s). The constants L_s in (8) are local post-layer ℓ_∞ -Lipschitz bounds; for ViTs this paper treats them as architecture-dependent and uses the empirical proxy audit in Appendix H.2.

G Gaussian Specialization and Spectral Interpretation

The deterministic mask certificates in Section 5 are closest to the implemented unstructured pruning operation; this appendix gives Gaussian sufficient conditions and spectral interpretations for admissible random-like components.

G.1 Assumptions for the Gaussian specialization

The finite- T convention from Section 4 remains in force unless stated otherwise.

Whenever λ_+ denotes the right MP edge for the eigenvalues of

$$X_\ell = \frac{1}{N} W_\ell^\top W_\ell,$$

we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding edge on the singular-value scale.

Assumption G.1 (Architecture and outer-layer scaling). We consider the simplified three-layer network

$$X(\alpha, s) = \lambda \circ W_3 \circ \lambda \circ W_2 \circ \lambda \circ W_1 s, \quad s \in T, \tag{9}$$

$$\phi(\alpha, s) = \rho \circ X(\alpha, s), \tag{10}$$

where λ is the coordinatewise absolute value or ReLU, and W_1, W_2, W_3 are the weight matrices. Biases are omitted for simplicity.

For fixed training time t , define

$$a(N, s) := \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}}. \quad (11)$$

We assume that for every $s \in T$, $a(N, s) \rightarrow 0$ as $N \rightarrow \infty$. The same assumption is imposed at any admissible limit point considered below.

Remark G.2 (Interpretation of Assumption G.1). This is the concrete version of the abstract factorization Assumption 4.1. The target layer is W_2 ; the computations before W_2 play the role of ψ_1 , and the computations after W_2 play the role of ψ_2 . The quantity $a(N, s)$ is the Gaussian-model version of $a_\phi(N, s) = h_\phi(s)d_1(N)$: it is the logit perturbation scale predicted by multiplying an $N^{-1/2}\sqrt{\log N}$ -size random effect through the outer layer. The assumption $a(N, s) \rightarrow 0$ is therefore the concrete data-path smallness condition needed for the loss and margin bounds.

Assumption G.3 (Admissible Gaussian decomposition of the middle layer). For each training time t , the middle layer $W_2(t) \in \mathbb{R}^{N \times N}$ admits a decomposition

$$W_2(t) = R_2(t) + S_2(t),$$

where:

1. $R_2(t)$ has i.i.d. centered Gaussian entries

$$R_2(t)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g(t)}{N}\right), \quad 0 \leq g(t) \leq 1;$$

where $g(t)$ is a deterministic variance scale;

2. $R_2(t)$ is independent of $S_2(t)$;
3. $R_2(t)$ is also independent of the outer-layer parameters $W_1(t)$ and $W_3(t)$.

Remark G.4. The variance scaling $g(t)/N$ ensures that $R_2(t)$ has bounded operator norm as $N \rightarrow \infty$. Rectangular variants $W_2(t) \in \mathbb{R}^{N \times M}$ with $M/N \rightarrow c \in (0, \infty)$ can be treated similarly.

Remark G.5 (Interpretation of Assumption G.3). This is a stylized additive-noise model, not a claim that the implemented ViT mask literally samples an independent Gaussian matrix. It is used to prove a sufficient condition: if the removed component behaves like an independent variance- g/N random matrix along the data path, then it satisfies the perturbation bounds required by the abstract theory. In the notation of Assumption 4.2, it gives $J(v) = \|v\|_2$, $d_1(N) \asymp \sqrt{\log N/N}$, and $d_2(N) \asymp 1/N$. The same variance scale has MP singular-value edge $2\sqrt{g}$ in the square case, which is why the fitted MP edge is a natural diagnostic in the algorithms.

Assumption G.6 (Sub-root- N structured component for the loss theorems). For each training time t , the structured component $S_2(t)$ satisfies

$$\|S_2(t)\|_F \leq B_N, \quad B_N = o(\sqrt{N}),$$

for some positive deterministic sequence B_N independent of t . Equivalently, the normalized size condition $\|S_2(t)\|_F^2/N \rightarrow 0$ is enough. The loss-reduction theorems use this assumption only to prove

$$\langle S_2(t), R_2(t) \rangle_F = o_{\mathbb{P}}(1);$$

one may replace it directly by this cross-term condition whenever it is known by another argument.

Remark G.7 (Interpretation of Assumption G.6). This assumption is a convenient Gaussian sufficient condition for the cross-term bound $\langle S_2, R_2 \rangle_F = o_{\mathbb{P}}(1)$. Conditional on S_2 ,

$$\text{Var}(\langle S_2, R_2 \rangle_F | S_2) = \frac{g(t)}{N} \|S_2\|_F^2.$$

Thus $B_N = o(\sqrt{N})$ forces this variance to vanish; this is the Gaussian counterpart of the explicit cross-term hypothesis used in Section 4.

Assumption G.8 (Approximate low rank for the spectral corollaries). Whenever a spectral conclusion is invoked, we additionally assume that the limit structured component $S_2^*(N)$ is approximately low rank in the sense that there exists a sequence $k_N = o(N)$ such that

$$\sum_{i > k_N} s_i(S_2^*(N))^2 \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

The exact fixed-rank regime is the special case $k_N \equiv r$.

Remark G.9 (Interpretation of Assumption G.8). This is the Gaussian specialization of Assumption 4.3 and is used only by the spectral corollaries.

Remark G.10 (No spectral-separation assumption). Assumptions G.6–G.8 do *not* require the informative singular values of S_2 to lie above the MP edge σ_+ . Thus signal directions may still be embedded in the bulk created by R_2 .

Definition G.11 (Admissible decomposition). A decomposition $W_2 = S_2 + R_2$ is called *admissible* if it satisfies Assumptions G.3–G.6. At a limit point $W_2^*(N)$, an admissible decomposition $W_2^*(N) = S_2^*(N) + R_2^*(N)$ additionally requires that the outer-layer scaling condition from Assumption G.1 holds for the corresponding limit outer layers $W_1^*(N), W_3^*(N)$, i.e.

$$a_*(N, s) := \|W_3^*(N)\|_{\infty} \|W_1^*(N)s\|_2 \sqrt{\frac{\log(2N)}{N}} \rightarrow 0 \quad \text{for each } s \in T.$$

A *spectrally admissible* decomposition is an admissible decomposition that also satisfies Assumption G.8.

Definition G.12 (Local-path convergence of an admissible decomposition). An admissible path decomposition

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

is said to converge locally to an admissible limit decomposition

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

if, for each fixed N ,

$$\|S_2(t, N) - S_2^*(N)\|_F + \|R_2(t, N) - R_2^*(N)\|_F \rightarrow 0 \quad \text{in probability as } t \rightarrow \infty.$$

G.2 Key perturbation lemma

Lemma G.13. Suppose Assumptions G.1 and G.3 hold. Define

$$\tau_N(s) := 2\sqrt{2} \|W_3\|_{\infty} \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2} a(N, s).$$

If we replace $W_2 = S_2 + R_2$ by S_2 , then

$$\mathbb{P}(\|X(s, \alpha_{S_2}) - X(s, \alpha_{W_2})\|_{\infty} \leq \tau_N(s)) \geq 1 - \frac{1}{N}.$$

Proof. See Appendix I, Proof of Lemma G.13.

For later use, define the classification margin on a finite test set T' :

$$\delta X(s, \alpha) := X_{C(s)}(s, \alpha) - \max_{j \neq C(s)} X_j(s, \alpha),$$

and the corresponding accuracy

$$\text{acc}_\alpha(T') := \frac{1}{|T'|} \#\{s \in T' : \delta X(s, \alpha) > 0\}.$$

Corollary G.14 (Margin-sensitive label stability). *Suppose Assumptions G.1 and G.3 hold. Let T' be a finite test set. Then*

$$\mathbb{P}\left(\left|\text{acc}_{\alpha_{W_2}}(T') - \text{acc}_{\alpha_{S_2}}(T')\right| \leq \frac{1}{|T'|} \#\left\{s \in T' : |\delta X(s, \alpha_{W_2})| \leq 2\tau_N(s)\right\}\right) \geq 1 - \frac{|T'|}{N}.$$

In particular, if

$$\min_{s \in T'} |\delta X(s, \alpha_{W_2})| > 2 \max_{s \in T'} \tau_N(s),$$

then

$$\text{acc}_{\alpha_{W_2}}(T') = \text{acc}_{\alpha_{S_2}}(T')$$

with probability at least $1 - \frac{|T'|}{N}$.

Proof. See Appendix J, Proof of Corollary G.14.

G.3 A theorem on how removing randomness reduces the regularized loss

We work with the ℓ_2 -regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha(t))) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i(t)\|_F^2, \quad \lambda_{\text{reg}} > 0. \quad (12)$$

We first isolate the unregularized perturbation term.

Lemma G.15. *Suppose Assumptions G.1–G.3 hold. Define the unregularized cross-entropy*

$$L_{\text{reg}}(\alpha) := -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha)).$$

Then there exists N_0 such that for all $N \geq N_0$,

$$\mathbb{P}\left(\left|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})\right| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N}.$$

In particular,

$$L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2}) \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Proof. See Appendix I, Proof of Lemma G.15.

Theorem G.16 (Confidence-parameter finite-sample perturbation bound). *Suppose Assumptions G.1 and G.3 hold, and fix a finite set T_0 and $\delta \in (0, 1)$. Define*

$$\tau_{N,\delta}(s) := \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{2g(t) \log(2N|T_0|/\delta)}{N}}.$$

Then, with probability at least $1 - \delta$,

$$\|X(s, \alpha_{S_2}) - X(s, \alpha_{W_2})\|_\infty \leq \tau_{N,\delta}(s) \quad \forall s \in T_0.$$

Consequently, for $T_0 = T$,

$$|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{2}{|T|} \sum_{s \in T} \tau_{N,\delta}(s)$$

with probability at least $1 - \delta$. If Assumption G.6 also holds, then for every $\eta > 0$,

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + \frac{2}{|T|} \sum_{s \in T} \tau_{N,\delta}(s) + 2\lambda_{\text{reg}}\eta\right) \geq 1 - \delta - 2e^{-N\eta^2/(2B_N^2)}.$$

Proof. See Appendix I, Proof of Theorem G.16.

Corollary G.17 (Growing-set Gaussian loss bound). *Suppose Assumptions G.1–G.3 hold with a finite labeled set T_N that may depend on N , and let $\delta_N \in (0, 1)$ with $\delta_N \rightarrow 0$. Let L_{reg, T_N} denote cross-entropy averaged over T_N , and let L_{T_N} denote the corresponding regularized objective with the same Frobenius penalty as in (12). Define*

$$b_{N,\delta_N}(T_N) := \frac{2}{|T_N|} \sum_{s \in T_N} \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{2g(t) \log(2N|T_N|/\delta_N)}{N}}.$$

If $b_{N,\delta_N}(T_N) \rightarrow 0$ in probability, then the cross-entropy loss averaged over T_N satisfies

$$L_{\text{reg}, T_N}(\alpha_{S_2}) - L_{\text{reg}, T_N}(\alpha_{W_2}) \rightarrow 0 \quad \text{in probability.}$$

If Assumption G.6 also holds, then

$$L_{T_N}(\alpha_{S_2}) \leq L_{T_N}(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + o_{\mathbb{P}}(1).$$

Proof. The bound follows by applying Theorem G.16 to the finite set T_N with failure probability δ_N ; the stated assumptions $\delta_N \rightarrow 0$ and $b_{N,\delta_N}(T_N) \rightarrow 0$ give the result.

Theorem G.18. *Suppose Assumptions G.1–G.6 hold. If $W_2 = R_2 + S_2$ and we replace W_2 by S_2 , then for every fixed training time t and every $\eta > 0$,*

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s) + 2\lambda_{\text{reg}}\eta\right) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}.$$

In particular,

$$L(\alpha_{S_2}) = L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

Proof. See Appendix J, Proof of Theorem G.18.

Corollary G.19. *Suppose Assumptions G.1–G.6 hold. For any $\vartheta \in [0, 1]$, define*

$$W_2(\vartheta) = S_2 + \vartheta R_2.$$

Then for every fixed training time t and every $\eta > 0$, there is an event $E_{N,\eta}$ with

$$\mathbb{P}(E_{N,\eta}) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}$$

such that on $E_{N,\eta}$, for every $\vartheta \in [0, 1]$,

$$L(\alpha_{W_2(\vartheta)}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R_2\|_F^2 + \frac{4\sqrt{2}(1 - \vartheta)}{|T|} \sum_{s \in T} a(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta.$$

In particular, for each fixed $\vartheta \in [0, 1]$,

$$L(\alpha_{W_2(\vartheta)}) = L(\alpha_{W_2}) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R_2\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0, 1]$.

Proof. See Appendix J, Proof of Corollary G.19.

Theorem G.20 (Entrywise thresholding in a sparse signal-plus-noise model). *Let $W_N = S_N + R_N \in \mathbb{R}^{N \times M_N}$, where R_{ij} are independent $N(0, g_N/N)$ random variables, $g_N \geq 0$, and S_N is deterministic or independent of R_N . Fix $\delta \in (0, 1)$ and set*

$$t_{N,\delta} := \sqrt{\frac{2g_N \log(2NM_N/\delta)}{N}}.$$

Define the entrywise hard-thresholded matrix

$$\mathcal{H}_t(W)_{ij} := W_{ij} \mathbf{1}_{\{|W_{ij}| > t\}}.$$

Then, with probability at least $1 - \delta$,

$$\{(i, j) : |(S_N)_{ij}| > 2t_{N,\delta}\} \subseteq \text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) \subseteq \text{supp}(S_N).$$

Equivalently, thresholding creates no false positives and can miss only entries of S_N whose magnitude is at most $2t_{N,\delta}$:

$$\text{supp}(S_N) \setminus \text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) \subseteq \{(i, j) : 0 < |(S_N)_{ij}| \leq 2t_{N,\delta}\}.$$

Let $\mathcal{B}_{N,\delta}$ be the beta-min event

$$\mathcal{B}_{N,\delta} := \left\{ \min_{(i,j) \in \text{supp}(S_N)} |(S_N)_{ij}| > 2t_{N,\delta} \right\},$$

with the condition interpreted as vacuous if $\text{supp}(S_N) = \emptyset$. If $\mathbb{P}(\mathcal{B}_{N,\delta}) \geq 1 - \gamma$, then entrywise thresholding W_N at level $t_{N,\delta}$ recovers the support of S_N exactly with probability at least $1 - \delta - \gamma$:

$$\text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) = \text{supp}(S_N).$$

In particular, if $\mathcal{B}_{N,\delta}$ holds deterministically, or almost surely when S_N is random, the recovery probability is at least $1 - \delta$. This result is a stylized support-recovery statement. Without beta-min separation it still gives a no-false-positive and large-signal-retention certificate, but exact support recovery requires all nonzero structured entries to clear the $2t_{N,\delta}$ noise buffer. It is not a proof that the ViT pruning masks identify the true structured support.

Proof. See Appendix I, Proof of Theorem G.20.

G.4 Asymptotic magnitude of admissible noise

We now replace exact local minimality by a weaker one-sided stationarity condition along the denoising direction $R_2^*(N)$.

Theorem G.21. *Suppose Assumptions G.1–G.6 hold along training. Assume that:*

1. *for each N , the parameter vector $\alpha(t, N)$ converges in probability as $t \rightarrow \infty$ to a limit $\alpha^*(N)$;*
2. *the limiting middle layer $W_2^*(N)$ admits an admissible decomposition*

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

in the sense of Definition G.11.

For $\vartheta \in [0, 1]$, let $\alpha_\vartheta^(N)$ denote the parameter vector obtained from $\alpha^*(N)$ by replacing $W_2^*(N)$ with $S_2^*(N) + \vartheta R_2^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event*

$$\liminf_{h \downarrow 0} \frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \geq -\epsilon_N$$

has probability tending to one. Define

$$\bar{a}_N := \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a_*(N, s).$$

Then

$$\|R_2^*(N)\|_F^2 \leq \frac{\bar{a}_N + \epsilon_N}{2\lambda_{\text{reg}}} + o_{\mathbb{P}}(1).$$

In particular, if $\bar{a}_N + \epsilon_N \rightarrow 0$ in probability, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \mathbb{P}(\|R_2^*(N)\|_F \leq \epsilon) = 1.$$

If, in addition, there exists an admissible path decomposition

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

which converges locally to the limit decomposition in the sense of Definition G.12, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{P}(\|R_2(t, N)\|_F \leq \epsilon) = 1.$$

Proof. See Appendix J, Proof of Theorem G.21.

Remark G.22. Theorem G.21 requires only one-sided directional stationarity at $\vartheta = 1$; a genuine local minimum gives $\epsilon_N \equiv 0$. Its second statement transfers the limit-point conclusion to any admissible training-path decomposition converging locally to the limit. Since the Gaussian $\|R_2^*(N)\|_F$ has no atom at any $\epsilon > 0$, pathwise convergence in probability yields convergence of the corresponding distribution functions at those thresholds.

Corollary G.23 (Variance-scale collapse at the limit point). *Under the hypotheses of Theorem G.21, write the admissible Gaussian limit perturbation as*

$$R_2^*(N)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

Here $g_*(N)$ is the deterministic variance scale of the limit perturbation. Then

$$g_*(N) N \rightarrow 0.$$

In particular, in the square case the MP singular-value edge of the admissible limit perturbation satisfies

$$\sigma_{+,*}^{\text{bulk}}(N) = 2\sqrt{g_*(N)} \rightarrow 0.$$

For a rectangular $N \times M_N$ perturbation with entries $N(0, g_*(N)/N)$, the corresponding Frobenius collapse condition is $g_*(N)M_N \rightarrow 0$; when $M_N/N \rightarrow c \in (0, \infty)$, this is equivalent to $g_*(N)N \rightarrow 0$.

Proof. See Appendix J, Proof of Corollary G.23.

Corollary G.24 (Bulk collapse and spike stabilization at the limit point). *Under the hypotheses of Theorem G.21, assume additionally that the limit decomposition is spectrally admissible in the sense of Definition G.11. Let*

$$s_1(W_2^*(N)) \geq \cdots \geq s_N(W_2^*(N))$$

denote the singular values of $W_2^(N)$, and let*

$$\sigma_1^*(N) \geq \sigma_2^*(N) \geq \cdots \geq 0$$

denote the singular values of $S_2^(N)$. Let $k_N = o(N)$ be as in Assumption G.8. Then:*

1. $\|R_2^*(N)\|_{\text{op}} \rightarrow 0$ in probability;

2. for each fixed $i \geq 1$,

$$|s_i(W_2^*(N)) - \sigma_i^*(N)| \rightarrow 0 \quad \text{in probability;}$$

3.

$$\sum_{i > k_N} s_i(W_2^*(N))^2 \rightarrow 0$$

in probability;

4. for every $\epsilon > 0$,

$$\nu_{W_2^*(N)}((\epsilon, \infty)) \rightarrow 0 \quad \text{in probability,}$$

where $\nu_{W_2^(N)}$ is the singular-value empirical spectral distribution.*

If, in addition, $\sigma_i^(N) \rightarrow \bar{\sigma}_i$ for some constants $\bar{\sigma}_i \geq 0$, then*

$$s_i(W_2^*(N)) \rightarrow \bar{\sigma}_i \quad \text{in probability for each fixed } i \geq 1.$$

Proof. See Appendix J, Proof of Corollary G.24.

Corollary G.25 (Singular-subspace stabilization under an asymptotic gap condition). *Under the hypotheses of Theorem G.21, let*

$$\sigma_1^*(N) \geq \sigma_2^*(N) \geq \cdots \geq 0$$

denote the singular values of $S_2^(N)$. Fix $k \geq 1$, and define*

$$\delta_k(N) := \sigma_k^*(N) - \sigma_{k+1}^*(N),$$

with the convention $\sigma_j^(N) = 0$ for j beyond the rank of $S_2^*(N)$. Assume that $\mathbb{P}(\delta_k(N) > 0) \rightarrow 1$ and that, after setting the ratio below to $+\infty$ on the event $\{\delta_k(N) = 0\}$,*

$$\frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)} \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Let $U_{*,k}(N)$ and $V_{*,k}(N)$ be orthonormal bases for the top- k left and right singular subspaces of $S_2^*(N)$, respectively. Let $\widehat{U}_k(N), \widehat{V}_k(N) \in \mathbb{R}^{N \times k}$ be matrices whose columns are orthonormal left and right singular vectors corresponding to the top k singular values of $W_2^*(N)$, and define the orthogonal projectors

$$\begin{aligned} P_{U_{*,k}}(N) &:= U_{*,k}(N)U_{*,k}(N)^\top, & P_{V_{*,k}}(N) &:= V_{*,k}(N)V_{*,k}(N)^\top, \\ \widehat{P}_{U,k}(N) &:= \widehat{U}_k(N)\widehat{U}_k(N)^\top, & \widehat{P}_{V,k}(N) &:= \widehat{V}_k(N)\widehat{V}_k(N)^\top. \end{aligned}$$

Then

$$\|\widehat{P}_{U,k}(N) - P_{U_{*,k}}(N)\|_{\text{op}} \rightarrow 0, \quad \|\widehat{P}_{V,k}(N) - P_{V_{*,k}}(N)\|_{\text{op}} \rightarrow 0$$

in probability as $N \rightarrow \infty$.

Proof. See Appendix J, Proof of Corollary G.25.

Remark G.26. Corollary G.25 converts spectral shrinkage into geometric stability: if residual operator-norm noise is asymptotically smaller than $\delta_k(N)$, the corresponding left/right singular subspaces become identifiable. This does not require a fixed spike floor; $\delta_k(N)$ may tend to 0 if it still dominates the noise.

Corollary G.27 (Eventual Gaussian supercriticality of persistent spikes). *Under the hypotheses of Theorem G.21, write*

$$R_2^*(N)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

Assume additionally that, for this corollary only, the admissible limit decomposition lies in the square Gaussian finite-rank deformation regime of Appendix J; in particular, $S_2^*(N)$ has rank at most r_0 for some deterministic r_0 independent of N . Assume that for some fixed $k \geq 1$ there exist constants

$$\bar{\sigma}_1 > \cdots > \bar{\sigma}_k > 0$$

such that

$$\sigma_i^*(N) \rightarrow \bar{\sigma}_i \quad \text{in probability as } N \rightarrow \infty$$

for each $1 \leq i \leq k$. Then:

1.

$$\mathbb{P}(\sigma_i^*(N) > \sqrt{g_*(N)}) \rightarrow 1 \quad \text{for each } 1 \leq i \leq k;$$

2. in the sense of Theorem J.3, these k spikes are eventually supercritical, so the corresponding singular values of $W_2^*(N)$ form BBP outlier clusters above the Gaussian bulk edge $2\sqrt{g_*(N)}$. Moreover, if the i -th spike is separated from the other nonzero singular values of $S_2^*(N)$ by a gap $\Delta_i(N) > 0$ with

$$\frac{\|R_2^*(N)\|_{\text{op}}}{\Delta_i(N)} \rightarrow 0 \quad \text{in probability,}$$

then the rank-one left and right spectral projectors of $W_2^*(N)$ associated with this spike converge in operator norm to the corresponding projectors of $S_2^*(N)$;

3. defining the inverse-BBP estimator on the event $s_i(W_2^*(N))^2 > 4g_*(N)$, whose probability tends to one, by

$$\widehat{\sigma}_i^{\text{BBP}}(N) := \frac{1}{2} \left(s_i(W_2^*(N)) + \sqrt{s_i(W_2^*(N))^2 - 4g_*(N)} \right),$$

and defining it arbitrarily off that event, one has

$$\widehat{\sigma}_i^{\text{BBP}}(N) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability}$$

for each $1 \leq i \leq k$.

Proof. See Appendix J, Proof of Corollary G.27.

Remark G.28. Corollary G.27 adds persistent-spike information to bulk collapse: a macroscopic $\bar{\sigma}_i > 0$ survives as $g_*(N) \rightarrow 0$, eventually exceeds the BBP threshold, and, under the stated gap condition, keeps its spectral direction; the inverse-BBP map then estimates the spike strength consistently.

Remark G.29 (Assumption map for the Gaussian section). Among the theorem-like results proved in the Gaussian section, the dependence on the hypotheses is as follows:

1. Theorem 2.2 and Proposition G.30 are standalone random-matrix statements and do not rely on Assumptions G.1–G.6.
2. Lemma G.13 and Corollary G.14 use only Assumptions G.1–G.3.
3. The first part of Theorem G.16 uses only Assumptions G.1–G.3.
4. Theorem G.18, Corollary G.19, and the regularized part of Theorem G.16 use Assumptions G.1–G.6; Assumption G.6 is used only to control $\langle S_2, R_2 \rangle_F$.
5. Theorem G.20 is a standalone sparse signal-plus-noise statement.
6. Theorem G.21 uses Assumptions G.1–G.6 together with the one-sided directional stationarity hypothesis stated in the theorem; its pathwise conclusion additionally uses Definition G.12.
7. Corollary G.23 is a consequence of Theorem G.21.
8. Corollary G.24 additionally uses Assumption G.8.
9. Corollary G.25 is a further consequence of Theorem G.21 plus the asymptotic gap condition

$$\|R_2^*(N)\|_{\text{op}}/\delta_k(N) \rightarrow 0 \quad \text{in probability.}$$

10. Corollary G.27 uses Theorem G.21 together with a persistent-spike assumption $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ and the additional exact finite-rank deformation hypothesis stated in the corollary itself.

Proposition G.30 (Spectral consequence of an MP-scale admissible random component). *Let $M_N/N \rightarrow c \in (0, \infty)$. Let*

$$W_N = R_N + S_N,$$

where $R_N \in \mathbb{R}^{N \times M_N}$ has i.i.d. $\mathcal{N}(0, g_N/N)$ entries with $g_N \rightarrow g_\infty > 0$, and S_N is independent of R_N with

$$\text{rank}(S_N) = o(\min\{N, M_N\}).$$

Then:

1. *the singular-value ESD ν_{R_N} , defined as in Definition 2.1 over the $\min\{N, M_N\}$ singular values of R_N , converges almost surely to the pushforward of the nonzero covariance-spectrum part of $\mu_{\text{MP}}^{c, g_\infty}$ under $x \mapsto \sqrt{x}$; equivalently, when $c > 1$ one first removes the MP atom at 0 and renormalizes. Its support is*

$$\left[\sqrt{g_\infty} |1 - \sqrt{c}|, \sqrt{g_\infty} (1 + \sqrt{c}) \right];$$

2. *the singular-value ESD of W_N has the same weak limit as ν_{R_N} , because*

$$\sup_{x \in \mathbb{R}} |\nu_{W_N}((-\infty, x]) - \nu_{R_N}((-\infty, x])| \leq \frac{\text{rank}(S_N)}{\min\{N, M_N\}} \rightarrow 0;$$

3. suppose, in addition, that S_N has fixed rank r , its nonzero singular values $\theta_i(N)$ converge to positive limits, and the pair (R_N, S_N) satisfies the finite-rank deformation hypotheses of Theorem J.4 with deterministic edge $b_+(N)$, critical threshold $\theta_c(N)$, and inverse- D outlier map. Then every spike for which $\mathbb{P}(\theta_i(N) > \theta_c(N) + \gamma) \rightarrow 1$ for some $\gamma > 0$ produces an outlier singular value $s_i(W_N)$ satisfying

$$s_i(W_N) - D_N^{-1} \left(\frac{1}{\theta_i(N)^2} \right) \rightarrow 0 \quad \text{in probability,}$$

and the inverse- D estimator from Theorem J.4 recovers $\theta_i(N)$ in probability. In the square Gaussian case $c = 1$, this reduces to the usual BBP outlier transition.

Thus any admissible random component that remains at MP scale leaves a persistent MP-type bulk in the singular spectrum.

Proof. See Appendix J, Proof of Proposition G.30.

In this stylized additive model, Theorem G.18 and Proposition G.30 say that an MP-scale R_2 leaves a persistent singular-value bulk, and removing it collapses that bulk while lowering the regularized objective by $\lambda_{\text{reg}} \|R_2\|_F^2$, up to the perturbation terms. Theorem G.21 further rules out admissible sub-MP Frobenius noise at locally stationary limit points, given local-path convergence.

H Supplementary Numerical Results

The supplementary numerical material covers MP-based pruning of fully connected DNNs on Fashion MNIST, outer-layer scaling diagnostics, and the noise-perturbation experiment. The extended version appears in the companion *methodology and extended-numerics* paper, [cast_2e/methodology.pdf](#) (LaTeX source: [cast_2e/methodology.tex](#)). The repository root is https://github.com/yspennstate/RMT_based_pruning_in_deep_learning.

H.1 Interpretation of the ViT-B/16 sweep

For the full layerwise interpretation of why RMT is used as a budget allocator rather than direct sub-edge singular-component removal, see the methodology paper, quick-reference key `sec:s33-interpretation` and the methodology paper, quick-reference key `sec:seb-protocols`.

H.2 Multi-architecture cert connection

Why the slope steepens with sparsity. The slope-steepening and bulk-reservoir interpretation of Figure 2 are discussed in the methodology paper, quick-reference key `sec:s34-arch-validation` and the methodology paper, quick-reference key `sec:s35-cert-numerics`.

Architecture and anomalous-cell provenance. Architecture provenance, adaptive-variant details, and the Hiera-Base+ anomalous-cell note are in the methodology paper, quick-reference key `sec:s34-arch-validation`; no correction was applied to the archived checkpoint.

Connecting the deterministic certificates to the multi-architecture numerics. The deterministic mask certificate (Theorem 5.4) certifies that pruning a removed component R from a layer $A = S + R$ decreases the elastic-net objective whenever the data-path budget satisfies

$$B_T(R) < \lambda_2 \|R\|_F^2 + \lambda_1 \|R\|_1, \quad (13)$$

where $B_T(R)$ is the data-path budget.

The full theory-to-numerics bridge, including the iid-Gaussian σ_+ caveat, prune–restore interpretation, operator-norm proxy audit, and 282-checkpoint correlation results, is given in the methodology paper, quick-reference key `sec:s35-cert-numerics`.

I Proofs of the perturbation lemmas

Lemma I.1 (Slack removal). *Let X_N and Y_N be real random variables. Suppose that for every fixed $\eta > 0$,*

$$(X_N - Y_N - \eta)_+ \rightarrow 0 \quad \text{in probability.}$$

Then

$$(X_N - Y_N)_+ \rightarrow 0 \quad \text{in probability.}$$

Equivalently, if $X_N \leq Y_N + \eta + o_{\mathbb{P}}(1)$ for every fixed $\eta > 0$, then $X_N \leq Y_N + o_{\mathbb{P}}(1)$.

Proof. Fix $\delta > 0$ and set $\eta = \delta/2$. Then

$$\mathbb{P}(X_N - Y_N > \delta) \leq \mathbb{P}(X_N - Y_N - \eta > \delta - \eta) \leq \mathbb{P}((X_N - Y_N - \eta)_+ > \delta/2),$$

and the last probability tends to 0 by assumption. $\square$

I.1 Proofs of the deterministic pruning certificates

Proof of Lemma 5.1. For each $s \in T$, the definition of L_s gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \leq \|\psi_2(A_\theta \psi_1(s)) - \psi_2(A_1 \psi_1(s))\|_\infty \leq (1 - \theta)L_s \|R\psi_1(s)\|_\infty.$$

For the single-sample cross-entropy

$$\ell_s(z) := -z_{C(s)} + \log \left(\sum_{j=1}^K e^{z_j} \right),$$

we have $\nabla \ell_s(z) = \rho(z) - e_{C(s)}$, hence $\|\nabla \ell_s(z)\|_1 \leq 2$. Therefore ℓ_s is 2-Lipschitz with respect to $\|\cdot\|_\infty$. Averaging over $s \in T$ yields

$$|\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1)| \leq \frac{2}{|T|} \sum_{s \in T} (1 - \theta)L_s \|R\psi_1(s)\|_\infty = (1 - \theta)B_T(R).$$

$\square$

Proof of Corollary 5.2. The logit perturbation bound in the proof of Lemma 5.1 gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \leq \eta_s(\theta).$$

If the predicted label changes, then the original top logit and some competing logit must be separated by at most $2\eta_s(\theta)$; otherwise every competing logit remains below the original top logit after perturbation. This proves the prediction-margin claim.

For accuracy, correctness is determined by the sign of the true-label margin. The true class logit can move by at most $\eta_s(\theta)$, and the maximum competing logit can move by at most $\eta_s(\theta)$, so the true-label margin can change by at most $2\eta_s(\theta)$. Therefore the sign of $\Delta_{A_1}^{\text{true}}(s)$ can change only if $|\Delta_{A_1}^{\text{true}}(s)| \leq 2\eta_s(\theta)$. Counting such samples gives the displayed bound. $\square$

Proof of Theorem 5.3. By Lemma 5.1,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \leq (1 - \theta)B_T(R).$$

For the Frobenius term,

$$\|S + \theta R\|_F^2 - \|S + R\|_F^2 = -(1 - \theta^2)\|R\|_F^2 - 2(1 - \theta)\langle S, R \rangle_F.$$

Multiplying this identity by λ_2 and adding the cross-entropy bound gives the displayed path inequality. The η -version follows from

$$-2\lambda_2(1 - \theta)\langle S, R \rangle_F \leq 2\lambda_2(1 - \theta)|\langle S, R \rangle_F|.$$

The full-removal sufficient condition is the case $\theta = 0$. □

Proof of Theorem 5.4. By Lemma 5.1,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \leq (1 - \theta)B_T(R).$$

Since $S \odot R = 0$,

$$\|S + \theta R\|_F^2 = \|S\|_F^2 + \theta^2\|R\|_F^2, \quad \|S + \theta R\|_1 = \|S\|_1 + \theta\|R\|_1.$$

Similarly,

$$\|A_1\|_F^2 = \|S\|_F^2 + \|R\|_F^2, \quad \|A_1\|_1 = \|S\|_1 + \|R\|_1.$$

Substituting these identities into $\mathcal{J}(A_\theta) - \mathcal{J}(A_1)$ gives

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - \lambda_1(1 - \theta)\|R\|_1.$$

Taking $\theta = 0$ proves the full-pruning statement and the sufficient decrease condition follows immediately. □

Proof of Theorem 5.5. Apply Theorem 5.4 to the mask decomposition

$$A = (S + Q) + P.$$

The pairwise disjoint-support assumption implies $(S + Q) \odot P = 0$, so the theorem applies with structured part $S + Q$ and removed part P . This gives the displayed path inequality and the full prune-restore statement by taking $\theta = 0$. The sufficient decrease condition follows immediately. □

Proof of Corollary 5.6. The all-pruned upper bound is Theorem 5.4 applied to

$$A = S + R, \quad R = Q + P.$$

The restored upper bound is Theorem 5.5. Since $Q \odot P = 0$,

$$\|R\|_F^2 = \|Q\|_F^2 + \|P\|_F^2, \quad \|R\|_1 = \|Q\|_1 + \|P\|_1.$$

Therefore the restored upper bound is smaller than the all-pruned upper bound exactly when

$$B_T(P) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 < B_T(R) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1,$$

which is equivalent to

$$B_T(R) - B_T(P) > \lambda_2\|Q\|_F^2 + \lambda_1\|Q\|_1.$$

□

Proof of Corollary E.1. The proof is the weighted coordinatewise version of Theorem 5.4. The same cross-entropy bound applies, and $S \odot R = 0$ gives

$$(S_{ij} + \theta R_{ij})^2 = (S_{ij})^2 + \theta^2 (R_{ij})^2, \quad |S_{ij} + \theta R_{ij}| = |S_{ij}| + \theta |R_{ij}|.$$

Multiplying by the nonnegative weights and summing yields the displayed inequality; $\theta = 0$ gives the full-pruning condition. $\square$

Proof of Corollary E.2. Set $\theta = 1 - h$ in Corollary E.1 and divide by $h > 0$. This gives

$$\frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \leq B_T(R) - (2-h) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Taking $h \downarrow 0$ and using one-sided stationarity yields

$$-\varepsilon \leq B_T(R) - 2 \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|,$$

which is the directional claim. If the pathwise lower bound holds for every θ , combine it with Corollary E.1 and divide by $1 - \theta$ to obtain the finite- θ inequality. $\square$

Proof of Theorem 5.7. This is Corollary E.2 with $\gamma_{2,ij} \equiv \lambda_2$ and $\gamma_{1,ij} \equiv \lambda_1$. $\square$

Proof of Theorem 5.8. Apply Theorem 5.4 with $\theta = 0$ at each pruning step in the network obtained after the previous $j - 1$ pruning operations. If $\alpha^{(j)}$ denotes the parameters after step j , then

$$\mathcal{J}(\alpha^{(j)}) - \mathcal{J}(\alpha^{(j-1)}) \leq -\lambda_{2,j} \|R_j\|_F^2 - \lambda_{1,j} \|R_j\|_1 + B_j(R_j).$$

Summing over $j = 1, \dots, M$ telescopes the left-hand side and proves the displayed multi-layer bound. $\square$

Proof of Corollary E.3. For each $s \in T$, telescoping the logits gives

$$\|z^{(M)}(s) - z^{(0)}(s)\|_\infty \leq \sum_{j=1}^M \|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \leq \eta_s^{\text{tot}}.$$

The cross-entropy bound follows from the 2-Lipschitz estimate above, and the prediction and accuracy bounds follow by the margin argument of Corollary 5.2 with $\eta_s(\theta)$ replaced by η_s^{tot} . $\square$

Proof of Theorem 5.9. Fix $h \in (0, 1]$, and let $\alpha_h^{(j)}$ be the network after the first j fractional pruning steps, with $\alpha_h^{(0)} = \alpha^{(0)}$ and $\alpha_h^{(M)} = \alpha_h$. At step j , apply the one-layer deterministic certificate with $\theta = 1 - h$ in the intermediate network $\alpha_h^{(j-1)}$. By the assumed validity of the fractional budget $B_j(R_j)$,

$$\mathcal{J}(\alpha_h^{(j)}) - \mathcal{J}(\alpha_h^{(j-1)}) \leq h B_j(R_j) - \lambda_{2,j} (2h - h^2) \|R_j\|_F^2 - \lambda_{1,j} h \|R_j\|_1.$$

Summing over $j = 1, \dots, M$ gives

$$\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)}) \leq h \sum_{j=1}^M B_j(R_j) - h(2-h) \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - h \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1.$$

After division by h , the one-sided directional stationarity assumption and the limit $h \downarrow 0$ imply

$$-\varepsilon \leq \sum_{j=1}^M B_j(R_j) - 2 \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1,$$

which is the first claim. If the stronger pathwise lower bound holds for every $h \in [0, 1]$, combine it with the displayed upper bound and divide by h to obtain the finite- h inequality. $\square$

Proof of Corollary 5.10. Condition on the vectors $v_s = \psi_1(s)$ and on the bounds L_s . For each $s \in T$ and row i ,

$$\langle r_i, v_s \rangle \sim N\left(0, \frac{g}{n} \|v_s\|_2^2\right).$$

Thus

$$\mathbb{P}\left(|\langle r_i, v_s \rangle| > \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}}\right) \leq \frac{\delta}{n|T|}.$$

A union bound over $i = 1, \dots, n$ and $s \in T$ gives, with probability at least $1 - \delta$,

$$\|Rv_s\|_\infty \leq \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}} \quad \forall s \in T.$$

Substitution into (8) proves the claim. $\square$

Proof of Corollary 5.11. For an $n \times m$ Gaussian matrix with entries $N(0, g/n)$, the nominal MP singular-value edge is

$$\sigma_{+,R} = \sqrt{g} \left(1 + \sqrt{\frac{m}{n}}\right).$$

Equivalently,

$$\sqrt{g} = \frac{\sigma_{+,R}}{1 + \sqrt{m/n}}.$$

Substituting this identity into the bound of Corollary 5.10 gives the displayed inequality. $\square$

Proof of Corollary E.5. Apply Corollary 5.10 at pruning step j with failure probability δ_j , conditional on the intermediate vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. This gives the displayed bound for $B_j(R_j)$ with conditional probability at least $1 - \delta_j$. A union bound over $j = 1, \dots, M$ gives simultaneous validity with probability at least $1 - \sum_j \delta_j \geq 1 - \delta$. Substituting these simultaneous budget bounds into Theorem 5.8 proves the objective inequality. $\square$

Proof of Corollary E.6. The proof is identical to Corollary E.5, using Corollary E.4 in place of Corollary 5.10. $\square$

Proof of Corollary E.4. Condition on the variables with respect to which v_s and L_s are measurable. Apply the assumed tail bound with

$$t_s := \|v_s\|_2 \sqrt{\frac{g}{cn} \log \frac{2n|T|}{\delta}}.$$

The assumed tail bound and the same row/sample union bound as above give

$$\|Rv_s\|_\infty \leq t_s \quad \forall s \in T$$

with probability at least $1 - \delta$. Substitution into (8) proves the corollary. $\square$

I.2 Proof of Lemma G.13

Proof. Let

$$\nu := \lambda(W_1 s).$$

Since λ is either absolute value or ReLU, it is coordinatewise 1-Lipschitz and satisfies $\|\nu\|_2 \leq \|W_1 s\|_2$. Write

$$X(s, \alpha_{W_2}) = \lambda(W_3 \lambda((S_2 + R_2)\nu)), \quad X(s, \alpha_{S_2}) = \lambda(W_3 \lambda(S_2 \nu)).$$

Using the 1-Lipschitz property of λ and $\|Av\|_\infty \leq \|A\|_\infty \|v\|_\infty$, we obtain

$$\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\|_\infty \leq \|W_3\|_\infty \|R_2 \nu\|_\infty.$$

So it suffices to bound $\|R_2 \nu\|_\infty$.

By Assumption G.3, R_2 is independent of W_1 , hence independent of ν . Conditioned on ν , each coordinate $(R_2 \nu)_j$ is centered Gaussian with variance

$$\text{Var}((R_2 \nu)_j \mid \nu) = \frac{g(t)}{N} \|\nu\|_2^2 \leq \frac{\|W_1 s\|_2^2}{N}.$$

Set

$$u_N := 2\sqrt{2} \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}}.$$

If $\|W_1 s\|_2 = 0$, then $\nu = 0$ and the claim is immediate. Otherwise, the elementary Gaussian tail bound gives, conditionally on ν ,

$$\mathbb{P}(|(R_2 \nu)_j| > u_N \mid \nu) \leq 2 \exp\left(-\frac{u_N^2}{2\|W_1 s\|_2^2/N}\right) = 2(2N)^{-4}.$$

A union bound over $j = 1, \dots, N$ gives

$$\mathbb{P}\left(\|R_2 \nu\|_\infty > 2\sqrt{2} \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} \mid \nu\right) \leq \frac{1}{8N^3} \leq \frac{1}{N}.$$

The same bound holds unconditionally after averaging over ν . Multiplying by $\|W_3\|_\infty$ yields the desired estimate. $\square$

I.3 Proof of Lemma 4.4

Proof. Let $u := \psi_1(s)$. Since

$$X(s, \alpha_W) = \psi_2((R + S)u), \quad X(s, \alpha_S) = \psi_2(Su),$$

Assumption 4.1 gives

$$\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \leq L_{\psi_2}(s) \|Ru\|_\infty.$$

Applying Assumption 4.2 with $v = u$ and then averaging over the conditioning variables gives

$$\mathbb{P}(\|Ru\|_\infty > d_1(N)J(u)) \leq d_2(N).$$

Therefore

$$\mathbb{P}(\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \leq d_1(N)J(\psi_1(s))L_{\psi_2}(s)) \geq 1 - d_2(N),$$

which is the claim. $\square$

I.4 Justification of Example F.6

The standard sub-Gaussian tail bound gives absolute constants $c, C > 0$ such that, conditionally on S, ψ_1, ψ_2 ,

$$\mathbb{P}(|\langle r_j, v \rangle| > u \mid S, \psi_1, \psi_2) \leq 2 \exp\left(-c \frac{Nu^2}{\kappa^2 \|v\|_2^2}\right).$$

Taking

$$u = C\kappa \|v\|_2 \sqrt{\frac{\log(2N/\delta)}{N}}$$

with C large enough and applying a union bound over $j = 1, \dots, N$ yields the displayed ℓ_∞ concentration bound.

I.5 Proof of Lemma G.15

Proof. For $s \in T$, let $X(\alpha, s) \in \mathbb{R}^K$ denote the vector of pre-softmax logits. By Lemma G.13, for each fixed $s \in T$,

$$\mathbb{P}\left(\|X(\alpha_{S_2}, s) - X(\alpha_{W_2}, s)\|_\infty \leq \tau_N(s)\right) \geq 1 - \frac{1}{N},$$

where

$$\tau_N(s) = 2\sqrt{2} \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2} a(N, s).$$

Let $\ell_s(z) = -z_{C(s)} + \log \sum_j e^{z_j}$. As in the proof of Lemma 5.1, ℓ_s is 2-Lipschitz with respect to $\|\cdot\|_\infty$. Applying this with $z = X(\alpha_{S_2}, s)$ and $z' = X(\alpha_{W_2}, s)$, we get

$$|\ell_s(X(\alpha_{S_2}, s)) - \ell_s(X(\alpha_{W_2}, s))| \leq 4\sqrt{2} a(N, s)$$

on the high-probability event from Lemma G.13. Since T is finite, a union bound over $s \in T$ yields

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N},$$

which proves the lemma. $\square$

I.6 Proof of Theorem G.16

Proof. For each $s \in T_0$, set $\nu_s = \lambda(W_1 s)$. Applying the same conditional Gaussian row bound and union bound as in Lemma G.13, now over $s \in T_0$ and $j = 1, \dots, N$ with failure probability δ , gives the displayed simultaneous bound on $\|R_2 \nu_s\|_\infty$. Multiplying by $\|W_3\|_\infty$ yields the logit bound, and the cross-entropy bound follows from the 2-Lipschitz estimate for ℓ_s .

For the regularized statement, use the exact identity

$$L(\alpha_{S_2}) - L(\alpha_{W_2}) = L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 - 2\lambda_{\text{reg}} \langle S_2, R_2 \rangle_F.$$

Conditional on S_2 ,

$$\langle S_2, R_2 \rangle_F \sim \mathcal{N}\left(0, \frac{g(t)}{N} \|S_2\|_F^2\right),$$

and Assumption G.6 gives

$$\mathbb{P}(|\langle S_2, R_2 \rangle_F| > \eta \mid S_2) \leq 2e^{-N\eta^2/(2B_N^2)}.$$

Intersecting the logit event with $\{|\langle S_2, R_2 \rangle_F| \leq \eta\}$ gives the claimed probability and inequality. $\square$

I.7 Proof of Theorem G.20

Proof. Condition on S_N when it is random; the remaining probability is over R_N . If $g_N = 0$, then $R_N = 0$ almost surely and the claim is immediate. Assume $g_N > 0$. For every entry,

$$\mathbb{P}(|R_{ij}| > t_{N,\delta}) \leq 2 \exp\left(-\frac{Nt_{N,\delta}^2}{2g_N}\right) = \frac{\delta}{NM_N},$$

and a union bound gives

$$\mathbb{P}\left(\max_{i,j} |R_{ij}| \leq t_{N,\delta}\right) \geq 1 - \delta.$$

On this event, if $(i, j) \notin \text{supp}(S_N)$, then $|W_{ij}| = |R_{ij}| \leq t_{N,\delta}$, so $\mathcal{H}_{t_{N,\delta}}$ removes that entry. Thus thresholding creates no false positives. If $|(S_N)_{ij}| > 2t_{N,\delta}$, then

$$|W_{ij}| \geq |(S_N)_{ij}| - |R_{ij}| > 2t_{N,\delta} - t_{N,\delta} = t_{N,\delta},$$

so thresholding keeps that entry. This proves the support sandwich and the missed-support inclusion. On the beta-min event $\mathcal{B}_{N,\delta}$, every element of $\text{supp}(S_N)$ has magnitude larger than $2t_{N,\delta}$, so the support sandwich collapses to exact support recovery. Unconditionally, the exact-recovery failure probability is at most $\delta + \mathbb{P}(\mathcal{B}_{N,\delta}^c) \leq \delta + \gamma$. $\square$

J Low-rank deformations and proofs of the main theorems

J.1 External random-matrix inputs quoted in the manuscript

The Marchenko–Pastur theorem in Theorem 2.2 and the square Gaussian finite-rank deformation theorem in Theorem J.1 are used as standard random-matrix inputs. The statements are classical; see, for example, [34, 61] for Marchenko–Pastur theory and [2] for the Benaych–Georges–Nadakuditi outlier theorem. All paper-specific consequences stated after these inputs are proved below.

J.2 Finite-rank deformations of random matrices

Theorem J.1 (Benaych–Georges–Nadakuditi, square Gaussian case [2]). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, 1/N)$ entries and S_N is deterministic, or independent of R_N , with fixed rank r and nonzero singular values $\sigma_1 > \dots > \sigma_r > 0$ independent of N . Then the i -th largest singular value $s_i(W_N)$ converges almost surely to*

$$s_i(W_N) \rightarrow \begin{cases} \sigma_i + \sigma_i^{-1}, & \sigma_i > 1, \\ 2, & \sigma_i \leq 1. \end{cases}$$

In particular, subcritical spikes merge with the bulk edge 2.

Proof. This is a classical cited input; see the source note in Appendix J.1. The variance- g and varying-scale consequences are proved immediately below.

Corollary J.2 (Square Gaussian case with variance g/N). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, g/N)$ entries for some fixed $g > 0$, and S_N is deterministic, or independent of R_N , with fixed rank r and distinct nonzero singular values $\sigma_1 > \dots > \sigma_r > 0$. Then*

$$s_i(W_N) \rightarrow \begin{cases} \sigma_i + g/\sigma_i, & \sigma_i > \sqrt{g}, \\ 2\sqrt{g}, & \sigma_i \leq \sqrt{g}. \end{cases}$$

Proof. Write

$$W_N = \sqrt{g}(\tilde{R}_N + \tilde{S}_N), \quad \tilde{R}_N := R_N/\sqrt{g}, \quad \tilde{S}_N := S_N/\sqrt{g}.$$

Then $\tilde{R}_N$ has i.i.d. $N(0, 1/N)$ entries, and the nonzero singular values of $\tilde{S}_N$ are $\sigma_i/\sqrt{g}$. Apply Theorem J.1 and multiply by $\sqrt{g}$. $\square$

Theorem J.3 (Square Gaussian outlier map with varying spike and noise scales). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, g_N/N)$ entries and $g_N \rightarrow g_\infty \in [0, \infty)$. Assume S_N has fixed rank r . Write its nonzero singular values in decreasing order,*

$$\sigma_1(S_N) \geq \dots \geq \sigma_r(S_N) > 0,$$

and assume that for each fixed $1 \leq i \leq r$,

$$\sigma_i(S_N) \rightarrow \bar{\sigma}_i \in (0, \infty) \quad \text{in probability}$$

where $\bar{\sigma}_1 \geq \dots \geq \bar{\sigma}_r > 0$, and that S_N is independent of R_N . Then for each fixed $1 \leq i \leq r$,

$$s_i(W_N) \rightarrow \begin{cases} \bar{\sigma}_i + g_\infty/\bar{\sigma}_i, & \bar{\sigma}_i > \sqrt{g_\infty}, \\ 2\sqrt{g_\infty}, & \bar{\sigma}_i \leq \sqrt{g_\infty}. \end{cases} \quad \text{in probability.}$$

Moreover, when $g_\infty > 0$, the usual BGN alignment conclusions hold for simple separated spikes: supercritical spikes have asymptotically nonzero projection onto the corresponding signal directions, while subcritical spikes have vanishing projection. For repeated or clustered spikes, these statements are understood as subspace-alignment statements. When $g_\infty = 0$, any separated spike cluster has convergent spectral projectors under the Wedin gap condition $\|R_N\|_{\text{op}}/\text{gap}_N \rightarrow 0$.

Proof. Condition on S_N . First suppose $g_\infty > 0$. Then $g_N > 0$ for all large N , and after dividing by $\sqrt{g_N}$ one reduces to the unit-variance square Gaussian finite-rank deformation theorem of Benaych-Georges and Nadakuditi [2]. On events whose probability tends to one, the conditioned spike values are arbitrarily close to their deterministic limits; the BGN outlier map is continuous away from the threshold, and at the threshold both branches give the edge. Thus the conditional deterministic-spike result and Slutsky's theorem give the displayed convergence in probability. The limiting outlier map in the unit-variance scaling is

$$\theta \mapsto \theta + \theta^{-1},$$

so in the original scaling it becomes

$$\sigma \mapsto \sigma + \frac{g_\infty}{\sigma},$$

with critical threshold $\sqrt{g_\infty}$ and edge $2\sqrt{g_\infty}$. The singular-vector and spike-subspace alignment statements in the positive-noise case are exactly the corresponding supercritical/subcritical BGN conclusions.

Now suppose $g_\infty = 0$. Write $R_N = \sqrt{g_N} Z_N$, where Z_N has i.i.d. $N(0, 1/N)$ entries. By the standard Gaussian operator-norm bound, $\|Z_N\|_{\text{op}} = O_{\mathbb{P}}(1)$, hence

$$\|R_N\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Weyl's singular-value inequality gives, for each fixed i ,

$$|s_i(W_N) - s_i(S_N)| \leq \|R_N\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Since $s_i(S_N) \rightarrow \bar{\sigma}_i$, we obtain $s_i(W_N) \rightarrow \bar{\sigma}_i$, which is the asserted formula with $g_\infty = 0$. The stated projector convergence in the separated-cluster case follows from Wedin's theorem. $\square$

Theorem J.4 (Abstract inverse- D spike recovery). *Let $W_N = R_N + S_N$. Assume that for each fixed $1 \leq i \leq r$, the i -th nonzero singular value of S_N is $\theta_i(N) > 0$. Assume further that there exist deterministic quantities $b_+(N)$, $\theta_c(N)$, and a strictly decreasing map*

$$D_N : (b_+(N), \infty) \rightarrow (0, D_N(b_+(N)^+))$$

with the following properties:

1. *on supercritical spike sequences, meaning $\mathbb{P}(\theta_i(N) > \theta_c(N)) \rightarrow 1$, one has*

$$s_i(W_N) - \rho_N(\theta_i(N)) \rightarrow 0 \quad \text{in probability,}$$

where

$$\rho_N(\theta) := D_N^{-1}\left(\frac{1}{\theta^2}\right);$$

2. *on subcritical spike sequences, meaning $\mathbb{P}(\theta_i(N) \leq \theta_c(N)) \rightarrow 1$, one has*

$$s_i(W_N) - b_+(N) \rightarrow 0 \quad \text{in probability;}$$

3. *in the supercritical case, the singular vectors associated with $s_i(W_N)$ have asymptotically nonvanishing projection onto the corresponding singular directions of S_N ;*
4. *for each fixed supercritical spike under consideration, $\mathbb{P}(\theta_i(N) > \theta_c(N)) \rightarrow 1$, $\theta_i(N) \rightarrow \bar{\theta}_i \in (0, \infty)$, and there exist deterministic intervals*

$$I_{i,N} \subset (b_+(N), \infty)$$

and deterministic constants $L_{i,N} < \infty$ such that D_N is $L_{i,N}$ -Lipschitz on $I_{i,N}$ and

$$\begin{aligned} \mathbb{P}(\rho_N(\theta_i(N)) \in I_{i,N}) &\rightarrow 1, & \mathbb{P}(s_i(W_N) \in I_{i,N}) &\rightarrow 1, \\ L_{i,N}|s_i(W_N) - \rho_N(\theta_i(N))| &\rightarrow 0 & \text{in probability.} \end{aligned}$$

Define the inverse- D estimator on the event $s_i(W_N) > b_+(N)$, whose probability tends to one for every fixed supercritical spike under consideration, by

$$\hat{\theta}_i^{(D)}(N) := D_N(s_i(W_N))^{-1/2},$$

and define it arbitrarily off this event. Then for every fixed supercritical spike,

$$\hat{\theta}_i^{(D)}(N) - \theta_i(N) \rightarrow 0 \quad \text{in probability.}$$

Proof. On the event $\{s_i(W_N) \in I_{i,N}, \rho_N(\theta_i(N)) \in I_{i,N}\}$, whose probability tends to one, the Lipschitz assumption gives

$$|D_N(s_i(W_N)) - D_N(\rho_N(\theta_i(N)))| \leq L_{i,N}|s_i(W_N) - \rho_N(\theta_i(N))|.$$

Therefore

$$D_N(s_i(W_N)) = D_N(\rho_N(\theta_i(N))) + o_{\mathbb{P}}(1) = \frac{1}{\theta_i(N)^2} + o_{\mathbb{P}}(1).$$

Since $\theta_i(N) \rightarrow \bar{\theta}_i \in (0, \infty)$, the quantity $\theta_i(N)^{-2}$ stays in a compact subinterval of $(0, \infty)$ for all large N . Applying the continuous map $x \mapsto x^{-1/2}$ on that compact interval yields

$$\hat{\theta}_i^{(D)}(N) - \theta_i(N) \rightarrow 0 \quad \text{in probability.}$$

This is the inverse-outlier-map viewpoint underlying D -transform shrinkage methods such as OptShrink [40]. □

J.3 Proof of Proposition G.30

Proof. Part (i) is the MP law on the covariance scale together with the pushforward $x \mapsto \sqrt{x}$ to the singular-value scale. Because ν_{R_N} averages over only $\min\{N, M_N\}$ singular values, when $c > 1$ one removes the covariance-side zero atom of the full MP law before pushing forward and renormalizes the remaining mass. Part (ii) follows from the rank inequality for empirical spectral distributions: if A, B are matrices of the same dimensions, then

$$\sup_x |\nu_A((-\infty, x]) - \nu_B((-\infty, x])| \leq \frac{\text{rank}(A - B)}{\min\{N, M_N\}}.$$

Applying this with $A = W_N$, $B = R_N$, and $A - B = S_N$ gives the claimed bound. Part (iii) is the rectangular outlier principle encoded by the inverse- D transform in Theorem J.4; the square Gaussian BBP statement follows as the special case $c = 1$. $\square$

J.4 Proofs of Theorem G.18, Corollary G.19, Corollary G.14, and Theorem G.21 (Gaussian instances)

Proof of Theorem G.18. Conditional on $S_2(t)$, Assumption G.3 gives

$$\langle S_2(t), R_2(t) \rangle_F = \text{tr}(S_2(t)^\top R_2(t)) \sim \mathcal{N}\left(0, \frac{g(t)}{N} \|S_2(t)\|_F^2\right).$$

Since $\|S_2(t)\|_F \leq B_N$ and $g(t) \leq 1$, for every $\eta > 0$,

$$\mathbb{P}(|\langle S_2(t), R_2(t) \rangle_F| > \eta \mid S_2(t)) \leq 2e^{-N\eta^2/(2B_N^2)}. \quad (14)$$

Hence

$$\langle S_2(t), R_2(t) \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty. \quad (15)$$

Write

$$L(\alpha) = L_{\text{reg}}(\alpha) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i\|_F^2.$$

If $W_2 = R_2 + S_2$ and we replace W_2 by S_2 , then

$$\begin{aligned} L(\alpha_{S_2}) - L(\alpha_{W_2}) &= (L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})) - \lambda_{\text{reg}}(\|W_2\|_F^2 - \|S_2\|_F^2) \\ &= (L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})) - \lambda_{\text{reg}}\|R_2\|_F^2 - 2\lambda_{\text{reg}}\langle S_2, R_2 \rangle_F. \end{aligned} \quad (16)$$

By Lemma G.15,

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N}.$$

Intersecting this event with $|\langle S_2, R_2 \rangle_F| \leq \eta$, whose probability is bounded below by $1 - 2e^{-N\eta^2/(2B_N^2)}$, yields the finite- N inequality in the theorem. Letting $N \rightarrow \infty$ and using (15) gives the asymptotic expansion. $\square$

Proof of Corollary G.19. Let $W_2(\vartheta) = S_2 + \vartheta R_2$. Then

$$\begin{aligned} L(\alpha_{W_2(\vartheta)}) - L(\alpha_{W_2}) &= (L_{\text{reg}}(\alpha_{W_2(\vartheta)}) - L_{\text{reg}}(\alpha_{W_2})) \\ &\quad - \lambda_{\text{reg}}\left((1 - \vartheta^2)\|R_2\|_F^2 + 2(1 - \vartheta)\langle S_2, R_2 \rangle_F\right). \end{aligned} \quad (17)$$

For each $s \in T$, write $\nu_s = \lambda(W_1 s)$. Let E_N^{logit} be the event on which

$$\|W_3\|_\infty \|R_2 \nu_s\|_\infty \leq \tau_N(s) \quad \forall s \in T.$$

By the same samplewise Gaussian estimate as in Lemma G.13 and a union bound over T ,

$$\mathbb{P}(E_N^{\text{logit}}) \geq 1 - \frac{|T|}{N}.$$

On E_N^{logit} , for every $\vartheta \in [0, 1]$ and every $s \in T$,

$$\|X(s, \alpha_{W_2(\vartheta)}) - X(s, \alpha_{W_2})\|_\infty \leq (1 - \vartheta) \|W_3\|_\infty \|R_2 \nu_s\|_\infty \leq (1 - \vartheta) \tau_N(s).$$

The 2-Lipschitz property of the cross-entropy therefore gives, on the same event,

$$|L_{\text{reg}}(\alpha_{W_2(\vartheta)}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}(1 - \vartheta)}{|T|} \sum_{s \in T} a(N, s)$$

for every $\vartheta \in [0, 1]$. Combining E_N^{logit} with $|\langle S_2, R_2 \rangle_F| \leq \eta$ proves the finite- N bound. The asymptotic expansion follows from (17) and (15). Since the finite- N event is uniform in ϑ , the same expansion also holds for deterministic sequences $\vartheta_N \in [0, 1]$. $\square$

Proof of Corollary G.14. By Lemma G.13, for each fixed $s \in T'$,

$$\mathbb{P}(\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\|_\infty \leq \tau_N(s)) \geq 1 - \frac{1}{N}.$$

A union bound over $s \in T'$ gives simultaneous vector control with probability at least $1 - \frac{|T'|}{N}$. On that event,

$$|\delta X(s, \alpha_{W_2}) - \delta X(s, \alpha_{S_2})| \leq 2\tau_N(s) \quad \forall s \in T'.$$

Therefore labels can only change on samples with $|\delta X(s, \alpha_{W_2})| \leq 2\tau_N(s)$, which yields the claim. $\square$

Proof of Theorem G.21. Set

$$b_N := \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a_*(N, s) \rightarrow 0.$$

Choose a deterministic sequence $\eta_N \downarrow 0$ such that

$$\frac{N\eta_N^2}{B_N^2} \rightarrow \infty;$$

such a sequence exists because $B_N = o(\sqrt{N})$. By Corollary G.19 applied at the limit point, there exists an event E_N with $\mathbb{P}(E_N) \rightarrow 1$ such that on E_N , for every $\vartheta \in [0, 1]$,

$$L(\alpha_{\vartheta}^*(N)) - L(\alpha^*(N)) \leq -\lambda_{\text{reg}}(1 - \vartheta^2) \|R_2^*(N)\|_F^2 + (1 - \vartheta)b_N + 2\lambda_{\text{reg}}(1 - \vartheta)\eta_N$$

Write $\vartheta = 1 - h$. Dividing by $h > 0$ gives

$$\frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \leq -\lambda_{\text{reg}}(2 - h) \|R_2^*(N)\|_F^2 + b_N + 2\lambda_{\text{reg}}\eta_N$$

for every $h \in (0, 1]$. On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h \downarrow 0}$ yields

$$2\lambda_{\text{reg}} \|R_2^*(N)\|_F^2 \leq b_N + \epsilon_N + 2\lambda_{\text{reg}}\eta_N.$$

Therefore

$$\|R_2^*(N)\|_F^2 \leq \frac{b_N + \epsilon_N}{2\lambda_{\text{reg}}} + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \rightarrow 0$, this proves the quantitative estimate in the theorem. The vanishing-in-probability statement follows immediately when $b_N + \epsilon_N \rightarrow 0$ in probability.

Now assume the local-path convergence hypothesis from Definition G.12. Fix $\epsilon > 0$. For each fixed N ,

$$\|R_2(t, N)\|_F \rightarrow \|R_2^*(N)\|_F \quad \text{in probability as } t \rightarrow \infty$$

by the continuous mapping theorem. Since $R_2^*(N)$ is Gaussian (possibly degenerate if $g_*(N) = 0$), the random variable $\|R_2^*(N)\|_F$ has no atom at any positive ϵ . Hence

$$\mathbb{P}(\|R_2(t, N)\|_F \leq \epsilon) \rightarrow \mathbb{P}(\|R_2^*(N)\|_F \leq \epsilon) \quad \text{as } t \rightarrow \infty.$$

Letting $N \rightarrow \infty$ and using the limit-point statement proves the theorem. $\square$

Proof of Corollary G.23. Suppose, toward a contradiction, that there exist $\eta_0 > 0$ and a subsequence such that $g_*(N)N \geq \eta_0$. Since

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i,j=1}^N Z_{ij}^2, \quad Z_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, 1),$$

we have

$$\frac{\|R_2^*(N)\|_F^2}{g_*(N)N} = \frac{1}{N^2} \sum_{i,j=1}^N Z_{ij}^2 \rightarrow 1 \quad \text{in probability.}$$

Therefore

$$\mathbb{P}(\|R_2^*(N)\|_F^2 \geq \eta_0/2) \rightarrow 1,$$

contradicting Theorem G.21. Hence $g_*(N)N \rightarrow 0$. The edge statement follows from the square-case MP singular-value edge $2\sqrt{g_*(N)}$. In the rectangular $N \times M_N$ case,

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i=1}^N \sum_{j=1}^{M_N} Z_{ij}^2, \quad \frac{1}{NM_N} \sum_{i,j} Z_{ij}^2 \rightarrow 1$$

in probability. Thus $\|R_2^*(N)\|_F \rightarrow 0$ forces $g_*(N)M_N \rightarrow 0$, and if $M_N/N \rightarrow c \in (0, \infty)$ this is equivalent to $g_*(N)N \rightarrow 0$. $\square$

Proof of Corollary G.24. Since $\|R_2^*(N)\|_{\text{op}} \leq \|R_2^*(N)\|_F$, Theorem G.21 immediately gives

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

This proves (i).

For (ii), singular-value Weyl perturbation gives

$$|s_i(W_2^*(N)) - s_i(S_2^*(N))| \leq \|R_2^*(N)\|_{\text{op}} \quad \forall i.$$

Hence for each fixed $i \geq 1$,

$$|s_i(W_2^*(N)) - \sigma_i^*(N)| \leq \|R_2^*(N)\|_{\text{op}} \rightarrow 0$$

in probability.

Let $S_{2,\leq k_N}^*(N)$ denote the best rank- k_N approximation of $S_2^*(N)$. By the Eckart–Young theorem,

$$\begin{aligned} \sum_{i>k_N} s_i(W_2^*(N))^2 &= \inf_{\text{rank}(A)\leq k_N} \|W_2^*(N) - A\|_F^2 \leq \|W_2^*(N) - S_{2,\leq k_N}^*(N)\|_F^2 \\ &= \|R_2^*(N) + (S_2^*(N) - S_{2,\leq k_N}^*(N))\|_F^2 \leq 2\|R_2^*(N)\|_F^2 + 2 \sum_{i>k_N} \sigma_i^*(N)^2, \end{aligned}$$

which proves (iii) using Theorem G.21 and Assumption G.8.

For (iv), fix $\epsilon > 0$. Since $\|W_2^*(N)\|_F \leq \|S_2^*(N)\|_F + \|R_2^*(N)\|_F$, we have

$$\nu_{W_2^*(N)}((\epsilon, \infty)) \leq \frac{\|W_2^*(N)\|_F^2}{N\epsilon^2} \leq \frac{2B_N^2 + 2\|R_2^*(N)\|_F^2}{N\epsilon^2},$$

which converges to 0 in probability because $B_N = o(\sqrt{N})$ and $\|R_2^*(N)\|_F \rightarrow 0$ in probability. The final claim follows from (ii). $\square$

Proof of Corollary G.25. By Corollary G.24(i),

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Hence the event

$$F_N := \left\{ \|R_2^*(N)\|_{\text{op}} \leq \frac{\delta_k(N)}{4} \right\}$$

satisfies $\mathbb{P}(F_N) \rightarrow 1$.

On F_N , Weyl's singular-value inequality gives

$$s_k(W_2^*(N)) \geq \sigma_k^*(N) - \|R_2^*(N)\|_{\text{op}},$$

while

$$s_{k+1}(W_2^*(N)) \leq \sigma_{k+1}^*(N) + \|R_2^*(N)\|_{\text{op}},$$

with the convention $\sigma_j^*(N) = 0$ beyond the rank of $S_2^*(N)$. Therefore,

$$s_k(W_2^*(N)) - s_{k+1}(W_2^*(N)) \geq \delta_k(N) - 2\|R_2^*(N)\|_{\text{op}} \geq \frac{\delta_k(N)}{2}$$

on F_N . Thus on F_N , the top- k singular subspace of $W_2^*(N)$ is separated from the rest by a positive gap.

Applying Wedin's $\sin \Theta$ theorem for singular subspaces [62, 53] to

$$W_2^*(N) = S_2^*(N) + R_2^*(N),$$

with target rank k , yields a universal constant $C > 0$ such that on F_N ,

$$\|\hat{P}_{U,k}(N) - P_{U^*,k}(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)}, \quad \|\hat{P}_{V,k}(N) - P_{V^*,k}(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)}.$$

Since $\|R_2^*(N)\|_{\text{op}}/\delta_k(N) \rightarrow 0$ in probability and $\mathbb{P}(F_N) \rightarrow 1$, both projector differences converge to 0 in probability. $\square$

Proof of Corollary G.27. By Corollary G.23,

$$g_*(N)N \rightarrow 0,$$

hence $g_*(N) \rightarrow 0$. Since $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ in probability for each $1 \leq i \leq k$,

$$\mathbb{P}(\sigma_i^*(N) > \sqrt{g_*(N)}) \rightarrow 1,$$

which proves (i).

By Corollary G.24(i),

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Hence Weyl's inequality gives

$$s_i(W_2^*(N)) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability}$$

for each fixed $1 \leq i \leq k$. Since $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ and $2\sqrt{g_*(N)} \rightarrow 0$, these singular values lie above the Gaussian bulk edge $2\sqrt{g_*(N)}$ with probability tending to one. Equivalently, in the exact finite-rank Gaussian deformation language of Theorem J.3, the persistent spikes are eventually supercritical. If the additional gap condition in (ii) holds for a spike, Wedin's theorem applied to $W_2^*(N) = S_2^*(N) + R_2^*(N)$ gives, for the associated left and right rank-one projectors,

$$\|\hat{P}_i(N) - P_i^*(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\Delta_i(N)} \rightarrow 0 \quad \text{in probability,}$$

and similarly for the right projector. This proves the strengthened projector statement in (ii).

Set $y_N = s_i(W_2^*(N))$. The preceding Weyl estimate gives $y_N - \sigma_i^*(N) \rightarrow 0$ in probability, hence $y_N \rightarrow \bar{\sigma}_i > 0$. Also $g_*(N) \rightarrow 0$, so the event $y_N^2 > 4g_*(N)$ has probability tending to one. On this event,

$$0 \leq y_N - \hat{\sigma}_i^{\text{BBP}}(N) = \frac{1}{2} \left(y_N - \sqrt{y_N^2 - 4g_*(N)} \right) = \frac{2g_*(N)}{y_N + \sqrt{y_N^2 - 4g_*(N)}} \rightarrow 0$$

in probability. Therefore

$$\hat{\sigma}_i^{\text{BBP}}(N) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability,}$$

which proves (iii). □

J.5 Proofs of the abstract-generalization corollaries and theorems

The results below are the abstract additive-perturbation analogues of the Gaussian-instance proofs of Appendix J. Each is obtained by substituting the abstract logit-perturbation lemma (Lemma 4.4) for the Gaussian logit perturbation bound (Lemma G.13) and applying the abstract Assumptions 4.1–4.3 in place of Gaussian Assumptions G.3–G.6.

Proof of Theorem 4.5. For $s \in T$, let $X(\alpha, s)$ denote the vector of pre-softmax logits. By Lemma 4.4, for each fixed $s \in T$,

$$\mathbb{P}(\|X(\alpha_S, s) - X(\alpha_W, s)\|_{\infty} \leq a_{\phi}(N, s)) \geq 1 - d_2(N),$$

where $a_{\phi}(N, s) = h_{\phi}(s)d_1(N)$. The single-sample cross-entropy is 2-Lipschitz in $\|\cdot\|_{\infty}$, so on this event $|\ell_s(X(\alpha_S, s)) - \ell_s(X(\alpha_W, s))| \leq 2a_{\phi}(N, s)$. A union bound over $s \in T$ gives

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_S) - L_{\text{reg}}(\alpha_W)| \leq \frac{2}{|T|} \sum_{s \in T} a_{\phi}(N, s)\right) \geq 1 - |T|d_2(N).$$

Since $a_\phi(N, s) \rightarrow 0$, $L_{\text{reg}}(\alpha_S) - L_{\text{reg}}(\alpha_W) \rightarrow 0$ in probability. Combined with the exact regularizer identity $\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R \rangle_F = \|R\|_F^2 + o_{\mathbb{P}}(1)$, this gives $L(\alpha_S) = L(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1)$. $\square$

Proof of Corollary F.3. Let E_N^{logit} be the samplewise logit event $\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \leq a_\phi(N, s)$ for all $s \in T$. By Lemma 4.4 and a union bound, $\mathbb{P}(E_N^{\text{logit}}) \geq 1 - |T|d_2(N)$. On this event the samplewise perturbation argument with $(1 - \vartheta)R$ in place of R gives, for every $\vartheta \in [0, 1]$,

$$|L_{\text{reg}}(\alpha_{W(\vartheta)}) - L_{\text{reg}}(\alpha_W)| \leq (1 - \vartheta) \frac{2}{|T|} \sum_{s \in T} a_\phi(N, s).$$

Let $E_N^{\text{cross}}(\eta) := \{|\langle S, R \rangle_F| \leq \eta\}$ and $E_{N, \eta} := E_N^{\text{logit}} \cap E_N^{\text{cross}}(\eta)$. Then $\mathbb{P}(E_{N, \eta}) \geq 1 - |T|d_2(N) - \mathbb{P}(|\langle S, R \rangle_F| > \eta)$. On $E_{N, \eta}$, the exact regularizer identity $\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1 - \vartheta^2)\|R\|_F^2 + 2(1 - \vartheta)\langle S, R \rangle_F$ implies

$$L(\alpha_{W(\vartheta)}) \leq L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R\|_F^2 + (1 - \vartheta) \frac{2}{|T|} \sum_{s \in T} a_\phi(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta$$

for every $\vartheta \in [0, 1]$, which is exactly the claim. $\square$

Proof of Corollary F.4. Since $\theta_c(N) \rightarrow 0$ and $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ in probability, $\mathbb{P}(\sigma_i^*(N) > \theta_c(N)) \rightarrow 1$ for each $1 \leq i \leq k$, which proves (i). On the same high-probability event, the assumed BGN-type deformation regime places the i -th spike above the critical threshold, so the corresponding singular value cluster of $W^*(N)$ lies above the bulk edge. If the gap condition in (ii) holds for a spike, Wedin's theorem gives $\|\widehat{P}_i(N) - P_i^*(N)\|_{\text{op}} \leq C\|R^*(N)\|_{\text{op}}/\Delta_i(N) \rightarrow 0$ in probability, for the left rank-one projectors (and similarly for the right projectors), proving (ii). Applying Theorem J.4 with $\theta_i(N) = \sigma_i^*(N)$ then gives $\widehat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \rightarrow 0$ in probability for each fixed $1 \leq i \leq k$, which is (iii). $\square$

Proof of Corollary F.1. By Lemma 4.4 and a union bound over $s \in T_N$, with probability at least $1 - |T_N|d_2(N)$,

$$\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \leq a_\phi(N, s) \quad \forall s \in T_N.$$

On this event, the 2-Lipschitz property of cross-entropy gives

$$|L_{\text{reg}, T_N}(\alpha_S) - L_{\text{reg}, T_N}(\alpha_W)| \leq \frac{2}{|T_N|} \sum_{s \in T_N} a_\phi(N, s).$$

The right-hand side is $o_{\mathbb{P}}(1)$, and the event has probability tending to one because $|T_N|d_2(N) \rightarrow 0$. The regularized identity

$$\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R \rangle_F$$

and the cross-term hypothesis then yield

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1).$$

$\square$

Proof of Corollary F.2. Set $W(\vartheta) = S + \vartheta R$. Repeating the argument above with perturbation $(1 - \vartheta)R$ in place of R , we get

$$L_{\text{reg}}(\alpha_{W(\vartheta)}) - L_{\text{reg}}(\alpha_W) = o_{\mathbb{P}}(1).$$

Also,

$$\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1 - \vartheta^2)\|R\|_F^2 + 2(1 - \vartheta)\langle S, R \rangle_F = (1 - \vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

Hence

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

The same argument, or the uniform finite- N pathwise bound of Corollary F.3, gives the statement along any deterministic sequence $\vartheta_N \in [0, 1]$. $\square$

Proof of Theorem 4.6. Order the pruned layers as $\mathcal{P} = \{\ell_1, \dots, \ell_m\}$. Let $\alpha^{(j)}$ denote the network parameters after the first j replacements, so $\alpha^{(0)}$ is the original network and $\alpha^{(m)} = \alpha^{(|\mathcal{P}|)}$. By assumption, at stage j the hypotheses of Theorem 4.5 apply to layer ℓ_j , hence

$$L(\alpha^{(j)}) = L(\alpha^{(j-1)}) - \lambda_{\text{reg}}\|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1).$$

Summing these identities over $j = 1, \dots, m$ and using that $m = |\mathcal{P}|$ is fixed gives

$$L(\alpha^{(m)}) = L(\alpha^{(0)}) - \lambda_{\text{reg}} \sum_{j=1}^m \|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1),$$

which is the claimed finite-layer telescoping formula.

For the growing-layer version, use the finite- N single-layer bound on the intersection $\bigcap_{\ell \in \mathcal{P}_N} E_{\ell, N}$, whose probability tends to one by the assumed summability of the failure probabilities. Summing the exact loss identities over the ordered layers gives the accumulated regularization decrease $-\lambda_{\text{reg}} \sum_{\ell \in \mathcal{P}_N} \|R_{\ell}\|_F^2$ plus the sum of the cross-entropy perturbation errors and the accumulated cross term. The two summability hypotheses make these remainders $o_{\mathbb{P}}(1)$, yielding the same formula with $\mathcal{P}_N$. $\square$

Proof of Theorem 4.7. Let $b_N := (2/|T|) \sum_{s \in T} a_{\phi}(N, s) \rightarrow 0$. Since $\langle S^*(N), R^*(N) \rangle_F = o_{\mathbb{P}}(1)$, choose a deterministic sequence $\eta_N \downarrow 0$ such that $\mathbb{P}(|\langle S^*(N), R^*(N) \rangle_F| > \eta_N) \rightarrow 0$. By Corollary F.3 applied at the limit point, there is an event E_N with $\mathbb{P}(E_N) \rightarrow 1$ such that on E_N , for every $\vartheta \in [0, 1]$,

$$L(\alpha_{\vartheta}^*(N)) - L(\alpha^*(N)) \leq -\lambda_{\text{reg}}(1 - \vartheta^2)\|R^*(N)\|_F^2 + (1 - \vartheta)b_N + 2\lambda_{\text{reg}}(1 - \vartheta)\eta_N.$$

Write $\vartheta = 1 - h$ and divide by $h > 0$:

$$\frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \leq -\lambda_{\text{reg}}(2 - h)\|R^*(N)\|_F^2 + b_N + 2\lambda_{\text{reg}}\eta_N \quad \forall h \in (0, 1].$$

On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h \downarrow 0}$ yields $2\lambda_{\text{reg}}\|R^*(N)\|_F^2 \leq b_N + \epsilon_N + 2\lambda_{\text{reg}}\eta_N$, so

$$\|R^*(N)\|_F^2 \leq (b_N + \epsilon_N)/(2\lambda_{\text{reg}}) + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \rightarrow 0$, this proves the quantitative estimate; the vanishing-in-probability statement follows whenever $b_N + \epsilon_N \rightarrow 0$ in probability. The pathwise conclusion under the convergence/continuity hypotheses follows because $\|R(t, N)\|_F \rightarrow \|R^*(N)\|_F$ in probability and $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$. $\square$

K Algorithms for RMT-based pruning diagnostics

The BEMA edge estimator, spectral-fit test, and cycle-based pruning algorithms for the classical RMT baseline are specified in the companion methodology paper introduced above.

L Spectral Edge Budgeting and hybrid RMT pruning protocols

Operational SEB/SER protocols, four-regime strategy, layer-type extension, SEB final comparison, Haar-SV preprocessing sweep, Hybrid Magnitude-SER protocol, and checkpoint validation ledgers used in Table 2 are specified in the companion methodology paper.

The algorithm below is the SER step used after the exact-magnitude low-sparsity prefix in Hybrid Magnitude-SER.

Algorithm 1 Spectral Edge Reallocation (SER) algorithm. The step prunes weights, restores selected weights from the removed-weight reservoir, and uses MP-budget allocation in Hybrid Magnitude-SER.

Require: Trained network with weights $\{W_\ell\}_{\ell=1}^L$; target sparsity $s \in (0, 1)$; over-prune ratio $\rho \in (1, 1 + \epsilon]$; layerwise MP edge signal $\{\sigma_+(\ell)\}_{\ell=1}^L$; budget-allocation parameters (β, s_d, p) ; short FT schedule $\mathcal{T}_{\text{FT}}$.

- 1: **function** SER_PRUNE_RESTORE(network, $s, \rho, \sigma_+, \beta, s_d, p$)
 - 2: Compute layerwise weights $w_\ell(s) = \max(0.1, 1 + \beta d(s) z_\ell)$ where $z_\ell = (\sigma_+(\ell) - \bar{\sigma}_+)/\text{sd}(\sigma_+)$ and $d(s) = (1 - s/s_d)_+^p$. ▷ Section 3
 - 3: Score each parameter $|W_\ell|_{ij}/w_\ell(s)$ and prune the smallest $\rho \cdot s$ fraction globally. ▷ Over-prune
 - 4: Define the reservoir $\mathcal{R} = \{(\ell, i, j) : (W_\ell)_{ij} \text{ pruned}\}$.
 - 5: With kept weights frozen, run a short selection phase (one ImageNet-1k epoch at the same optimiser/LR/batch as the eventual fine-tune) that allows only the reservoir entries to receive gradients. Record the absolute movement $|\Delta_{ij}^{(\ell)}|$ of each reservoir entry from its pre-selection value.
 - 6: Rank reservoir entries by $|\Delta_{ij}^{(\ell)}| \cdot \rho_\ell(s)$ where $\rho_\ell(s) = w_\ell(s)$ is the layer’s σ_+ -derived budget weight from Step 1, recomputed at the post-prune sparsity. ▷ RMT-ranked reinsertion
 - 7: Restore the top entries until realised post-restoration sparsity equals s .
 - 8: Freeze the resulting zero mask M on all layers. Run $\mathcal{T}_{\text{FT}}$ (one fine-tuning epoch in our schedule) with $W_\ell \leftarrow W_\ell \odot M$ re-applied at each step.
 - 9: **return** restored network and final mask M .
 - 10: **end function**
-

M CAST: certificate-aware 2:4 + ToMe conversion

The original CAST 2:4+ToMe pipeline, generalized CAST-2E $k:n$ pipeline, permutation alignment, and inline-FT runner are specified in the companion methodology paper.

Justification of Algorithm 2. The cost $c_{i,g}(m)$ is the squared ℓ^2 form of the removed-mass contribution to output channel i of layer ℓ , evaluated on the calibration set. Lemma 5.1 bounds $|\Delta \text{CE}|$ through the literal quantity $L_s \|R\psi_1(s)\|_\infty$. CAST does not estimate L_s ; it minimises the calibration-set ℓ^2 surrogate $r^\top C_g r$ for $R = W - W'$, treating it as a certificate-inspired proxy under the activation distribution captured by C_g . The free-restoration step materialises the slot-value rule of Theorem 5.5: slots that the unstructured mask had zeroed but the chosen $k:n$ pattern keeps are filled from the dense checkpoint at zero sparse-execution FLOP cost, which can improve the certificate bound under the conditions of Corollary 5.6. The α_{ser} prior pulls the chosen pattern towards the RMT-allocated SER mask used in the ViT-B experiments.

Algorithm 2 CAST $k:n$ structured-projection algorithm. The projection selects row-wise structured masks; Stage 2 applies frozen-mask distillation.

Require: Linear layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with $d_{\text{in}} = nG$; SER source mask $M^{\text{SER}} \in \{0, 1\}^{d_{\text{out}} \times d_{\text{in}}}$ and SER weight W^{SER} ; dense pretrained W^{dense} ; calibration set $\mathcal{C}$ (row-specific size in Table 8); $k:n$ pattern; flag **free_restoration**; $\alpha_{\text{ser}} \geq 0$.

```

1: function CAST_PROJECT_KN( $W, M^{\text{SER}}, W^{\text{SER}}, W^{\text{dense}}, \mathcal{C}, k, n, \alpha_{\text{ser}}, \text{free\_restoration}$ )
2:   Reshape rows of  $W$  into  $G$  contiguous  $n$ -tuples; let  $W_{i,g,k}$  index the  $k$ -th weight of group  $g$  in
   output row  $i$ .
3:   For each input-channel group  $g$ , capture activations  $h_g(x) \in \mathbb{R}^n$  over  $\mathcal{C}$  and form within-group
   covariance  $C_g = \frac{1}{|\mathcal{C}|} \sum_{x \in \mathcal{C}} h_g(x) h_g(x)^\top$ .
4:   for each output row  $i = 1, \dots, d_{\text{out}}$  and group  $g = 1, \dots, G$  in parallel do
5:     for each candidate keep pattern  $m \in \{0, 1\}^n$  with  $|m|_0 = k$  do
6:       Define source tuple  $u_{i,g,k} = \begin{cases} W_{i,g,k}^{\text{SER}} & \text{if } M_{i,g,k}^{\text{SER}} = 1, \\ W_{i,g,k}^{\text{dense}} & \text{if } M_{i,g,k}^{\text{SER}} = 0 \text{ and } \text{free\_restoration}, \\ 0 & \text{otherwise.} \end{cases}$ 
7:       Materialise retained tuple  $v_{i,g,k}(m) = m_k \cdot u_{i,g,k}$ .
8:       Compute removed tuple  $r_{i,g}(m) = (1 - m) \odot u_{i,g}$   $\triangleright$  entries not retained
9:       Compute cost  $c_{i,g}(m) = r_{i,g}(m)^\top C_g r_{i,g}(m) + \alpha_{\text{ser}} \bar{c}_g |m - M_{i,g,:}^{\text{SER}}|_1$ ,
10:  where  $\bar{c}_g = \text{mean}_{i,k} u_{i,g,k}^2 \cdot \mathbb{E}_x[h_{g,k}(x)^2]$  normalises the prior to the cost magnitude.
11:     end for
12:      $m_{i,g}^* \leftarrow \arg \min_{m \in \mathcal{M}_{k:n}} c_{i,g}(m)$ .
13:     Set the new weight tuple  $W'_{i,g,:} \leftarrow v_{i,g,:}(m_{i,g}^*)$ .
14:   end for
15:   return projected layer  $W'$  and the per-row  $k:n$  mask  $M_{i,g,k}^{\text{CAST}} = m_{i,g}^*[k]$ .
16: end function

```
